# Student Workspace

## Offline-First Academic Productivity Engine with Embedded WebAssembly SQLite



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|---|---|
| **Student Author:** Abhishek Kumar | **Institution:** Chandigarh University |
| **Technology Stack:** WebAssembly SQLite (sql.js) • Next.js React • HTML5/CSS3 | **Dual Deployment:** GitHub Pages (Static) & Vercel (Serverless) |
| **Database Engine:** In-Browser SQLite with Local Binary Persistence | **Initial State:** Unseeded Clean Schema (User Controlled) |



## 1. Executive Summary & Problem Formulation

University engineering students frequently juggle demanding coursework, multiple lab deliverables, complex 10-point GPA calculations, and active exam preparation. Commercial software tools exacerbate cognitive fatigue through mandatory cloud accounts, recurring subscription fees, heavy telemetry tracking, and external database dependencies that fail when offline.

**Student Workspace** is an offline-first academic management suite built from the ground up to operate completely self-contained. By leveraging **WebAssembly-compiled SQLite (sql.js)**, the application brings full relational database capabilities—including ACID transactions, SQL queries, and structured schema integrity—directly into the student's browser. It supports seamless dual deployment on both **GitHub Pages** (via static root files) and **Vercel** (via Next.js serverless architecture).

| Student Pain Point | Commercial Limitations | Student Workspace Solution |
|---|---|---|
| Disorganized Coursework | Scattered across disconnected portals with missed deadlines. | Unified priority-ranked queue with due date countdowns. |
| Manual GPA Calculations | Error-prone manual credit multiplication across 10-point scales. | Automated SGPA calculator with honors classification and simulator. |
| Study Burnout & Procrastination | Unstructured marathon studying causing cognitive fatigue. | 25-minute Pomodoro cycles with synchronized breaks & streak logs. |
| Concept Retention Gaps | Passive reading of course notes yields poor exam recall. | Active recall flashcards with interactive 3D flip transitions. |
| Database & Server Fragility | Requires hosted cloud databases, API keys, and internet connectivity. | Local WebAssembly SQLite with binary persistence; 100% offline-ready. |

# 2. Functional Modules & Architecture

## Module 1: Assignments & Academic Deadlines Tracker

Provides a streamlined task queue designed specifically for university assignments, term papers, and laboratory deliverables. Each task record includes title, course code, due date, estimated effort hours, and priority categorization (High / Medium / Low). State changes trigger instant SQL UPDATE tasks SET completed = ? WHERE id = ? operations in the local SQLite engine.

## Module 2: SGPA Grade Calculator & Target Simulator

Implements the standard university 10-point credit grading system (A+ = 10.0 to F = 0.0). The engine calculates Semester Grade Point Average: $$\text{SGPA} = \frac{\sum (\text{Credits}_i \times \text{GradePoint}_i)}{\sum \text{Credits}_i}$$ and classifies academic standing into *First Class with Distinction* (SGPA ≥ 9.0), *First Class with Honors* (≥ 8.0), or *First Class* (≥ 6.5). The built-in Target Simulator calculates the exact average grade point required across upcoming modules to reach a desired target CGPA.

## Module 3: Pomodoro Focus Study Timer

Based on the Cirillo cognitive focus methodology, the timer enforces structured 25-minute study sprints with 5-minute short breaks and 15-minute long breaks. Features an animated SVG circular progress readout, interval state machine, and automatic logging of completed study blocks into the pomodoro_sessions SQL table.

## Module 4: Active Recall Revision Flashcards

Leverages cognitive psychology principles of retrieval practice. Each card features a 3D perspective flip (rotateY(180deg)) separating the concept question prompt from the detailed answer explanation. Includes keyboard navigation (Spacebar to flip) and mastery categorization.

# 3. Local SQLite Schema & WebAssembly Engine

The workspace initializes a relational schema across four dedicated tables in SQLite via WebAssembly (sql.js). Per product specifications, the schema initializes **clean and unseeded**, ensuring that students retain complete ownership over their workspace data:

```
-- 1. Courses Table
CREATE TABLE IF NOT EXISTS courses (
  id TEXT PRIMARY KEY,
  code TEXT NOT NULL,
  name TEXT NOT NULL,
  credits REAL NOT NULL,
  grade TEXT NOT NULL,
  grade_point REAL NOT NULL,
  created_at INTEGER NOT NULL
);

-- 2. Tasks Table
CREATE TABLE IF NOT EXISTS tasks (
  id TEXT PRIMARY KEY,
  title TEXT NOT NULL,
  course TEXT NOT NULL,
  due_date TEXT NOT NULL,
  priority TEXT NOT NULL,
  hours INTEGER NOT NULL,
  completed INTEGER NOT NULL DEFAULT 0,
  created_at INTEGER NOT NULL
);

-- 3. Flashcards Table
CREATE TABLE IF NOT EXISTS flashcards (
  id TEXT PRIMARY KEY,
  subject TEXT NOT NULL,
  question TEXT NOT NULL,
  answer TEXT NOT NULL,
  mastered INTEGER NOT NULL DEFAULT 0,
  created_at INTEGER NOT NULL
);

-- 4. Pomodoro Sessions Table
CREATE TABLE IF NOT EXISTS pomodoro_sessions (
  id TEXT PRIMARY KEY,
  subject TEXT NOT NULL,
  duration_minutes INTEGER NOT NULL,
  completed_at INTEGER NOT NULL
);
```

# 4. Dual Deployment Architecture

Student Workspace is uniquely designed to support dual zero-friction deployment targets:

- **GitHub Pages Deployment:** Hosted directly from the repository's root branch. The application executes WebAssembly SQLite (sql.js) entirely inside the client's browser, persisting exported database binaries to local storage. Configured with .nojekyll and an automated GitHub Actions deployment workflow (.github/workflows/pages.yml).
- **Vercel Deployment:** Fully compatible with Vercel's serverless edge infrastructure via Next.js (pages/index.js and pages/api/*). The sql.js engine runs seamlessly inside Vercel's Linux serverless functions without native compilation failures.
- **Local Preview:** Can be previewed instantly via any local static server (e.g., python3 -m http.server 3000) or via npm run dev.

# 5. Application Screenshots & Visual Verification

Below are authentic, high-resolution screenshots captured directly from the running web application of Student Workspace, demonstrating all core functional modules:

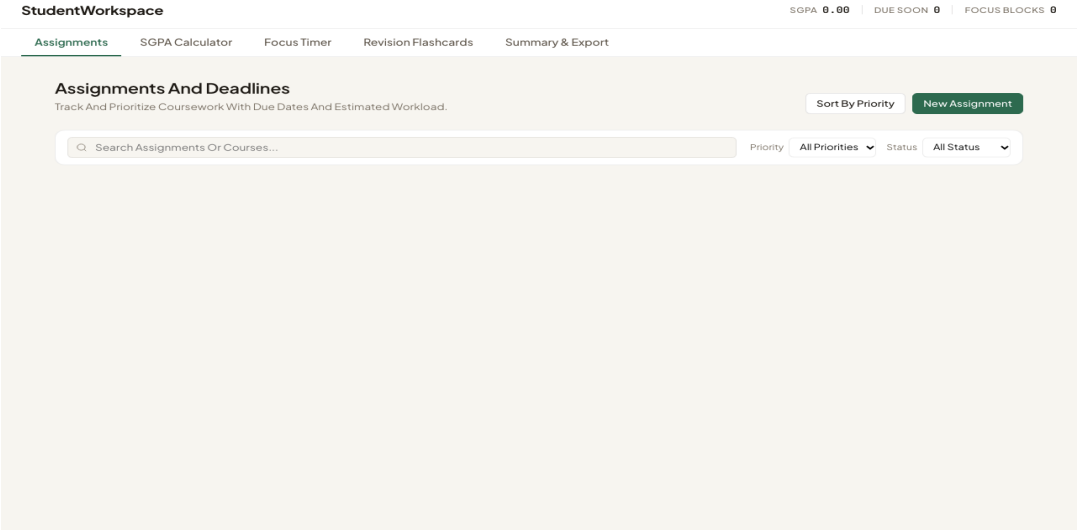



*Figure 1: Module 1 — Assignments & Deadlines Tracker with priority filtering and urgency badges.*

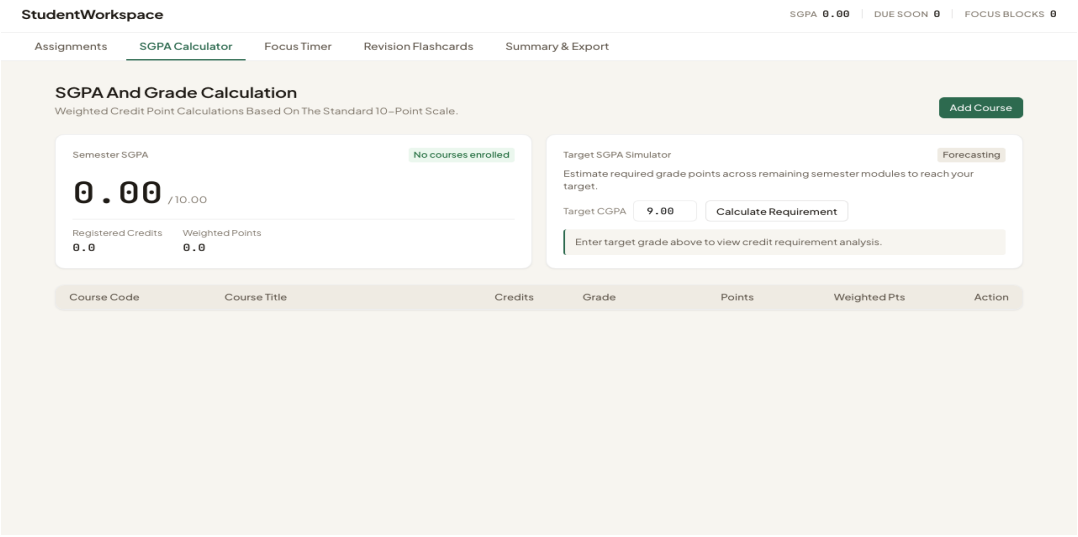



*Figure 2: Module 2 — Interactive SGPA & CGPA Grade Calculator with Distinction classification and target simulator.*

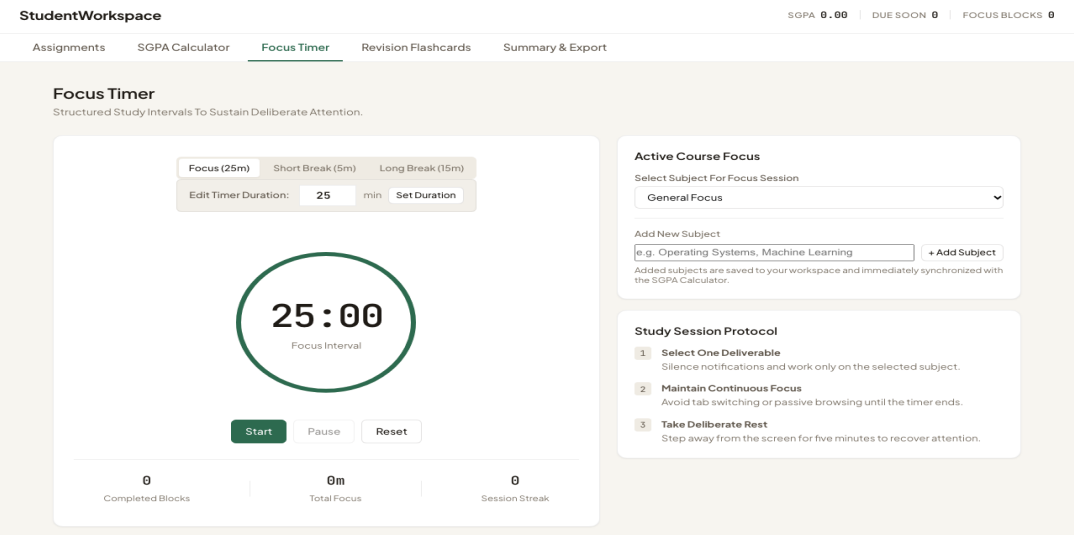



*Figure 3: Module 3 — Pomodoro Focus Study Timer with circular animated progress dial and session logging.*

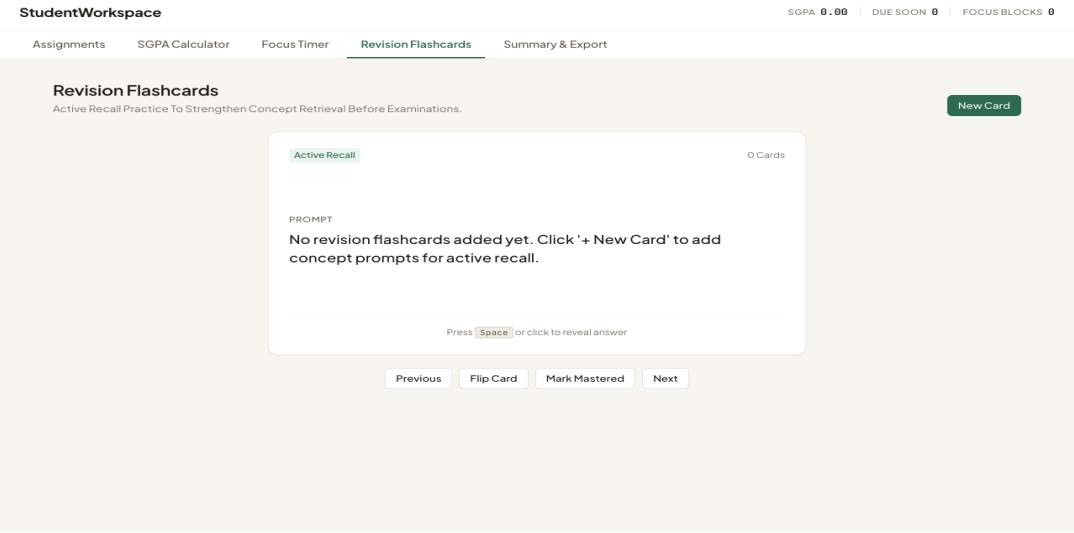



*Figure 4: Module 4 — Active Recall Revision Flashcards with 3D flip card interaction and mastery tracking.*

# 6. Complete Production Source Code

In accordance with project deliverables, all source code files powering Student Workspace are reproduced below verbatim. The codebase executes universally across modern web browsers without external database servers:

## 1. Markup: index.html (Semantic Structure, Modals & Navigation)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no">
  <title>Student Workspace</title>
  <meta name="description" content="A calm, offline-first academic productivity workspace powered by local SQLite. Fe
atures assignment tracking, SGPA/CGPA forecasting, Pomodoro focus timer, and active recall flashcards.">
  <link rel="stylesheet" href="style.css">
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700&family=JetBrains+Mono:w
ght@400;500;600&display=swap" rel="stylesheet">
  <!-- Client-side WebAssembly SQLite Engine -->
  <script src="https://cdnjs.cloudflare.com/ajax/libs/sql.js/1.8.0/sql-wasm.js"></script>
</head>
<body>
  <div class="app-layout">
    <!-- Top Header Bar -->
    <header class="app-header">
      <div class="header-left">
        <span class="brand-title">StudentWorkspace</span>
      </div>

      <div class="header-right">
        <div class="status-metric">
          <span class="metric-label">SGPA</span>
          <span class="metric-value" id="quickSgpa">0.00</span>
        </div>
        <div class="status-divider"></div>
        <div class="status-metric">
          <span class="metric-label">Due Soon</span>
          <span class="metric-value" id="quickPending">0</span>
        </div>
        <div class="status-divider"></div>
        <div class="status-metric">
          <span class="metric-label">Focus Blocks</span>
          <span class="metric-value" id="quickPomodoro">0</span>
        </div>
      </div>
    </header>

    <!-- Navigation Bar -->
    <nav class="app-nav">
      <div class="nav-container">
        <div class="nav-tabs">
          <button class="nav-tab active" data-tab="assignments">Assignments</button>
```

```
          <button class="nav-tab" data-tab="gpa">SGPA Calculator</button>
          <button class="nav-tab" data-tab="pomodoro">Focus Timer</button>
          <button class="nav-tab" data-tab="flashcards">Revision Flashcards</button>
          <button class="nav-tab" data-tab="analytics">Summary &amp;amp; Export</button>
        </div>
      </div>
    </nav>

    <!-- Main Content -->
    <main class="app-main">

      <!-- ==================== TAB 1: ASSIGNMENTS ==================== -->
      <section id="tab-assignments" class="tab-pane active">
        <div class="section-header">
          <div>
            <h1 class="section-title">Assignments And Deadlines</h1>
            <p class="section-desc">Track And Prioritize Coursework With Due Dates And Estimated Workload.</p>
          </div>
          <div class="section-actions">
            <button class="btn btn-secondary" id="btnSortUrgency">
              <span>Sort By Priority</span>
            </button>
            <button class="btn btn-primary" id="btnOpenTaskModal">
              <span>New Assignment</span>
            </button>
          </div>
        </div>

        <!-- Filter Bar -->
        <div class="filter-panel">
          <div class="search-field">
            <svg class="field-icon" width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" stro
ke-width="1.5"><circle cx="11" cy="11" r="8"/><line x1="21" y1="21" x2="16.65" y2="16.65"/></svg>
            <input type="text" id="taskSearchInput" placeholder="Search Assignments Or Courses...">
          </div>
          <div class="filter-controls">
            <div class="select-wrapper">
              <label class="select-label">Priority</label>
              <select id="priorityFilter" class="form-select-sm">
                <option value="ALL">All Priorities</option>
                <option value="High">High</option>
                <option value="Medium">Medium</option>
                <option value="Low">Low</option>
              </select>
            </div>
            <div class="select-wrapper">
```

```
            <label class="select-label">Status</label>
            <select id="statusFilter" class="form-select-sm">
              <option value="ALL">All Status</option>
              <option value="PENDING">Pending</option>
              <option value="COMPLETED">Completed</option>
            </select>
          </div>
        </div>
      </div>

      <!-- Task Grid -->
      <div class="task-grid" id="taskList">
        <!-- Dynamically populated -->
      </div>
    </section>

    <!-- ==================== TAB 2: GPA CALCULATOR ==================== -->
    <section id="tab-gpa" class="tab-pane">
      <div class="section-header">
        <div>
          <h1 class="section-title">SGPA And Grade Calculation</h1>
          <p class="section-desc">Weighted Credit Point Calculations Based On The Standard 10-Point Scale.</p>
        </div>
        <button class="btn btn-primary" id="btnOpenCourseModal">
          <span>Add Course</span>
        </button>
      </div>

      <div class="gpa-overview-grid">
        <!-- Summary Card -->
        <div class="card gpa-metric-card">
          <div class="metric-top">
            <span class="card-caption">Semester SGPA</span>
            <span class="status-tag tag-green" id="dispClassification">No courses enrolled</span>
          </div>
          <div class="gpa-display-row">
            <span class="gpa-large-number" id="dispSgpa">0.00</span>
            <span class="gpa-max-scale">/ 10.00</span>
          </div>
          <div class="gpa-breakdown-row">
            <div class="breakdown-item">
              <span class="breakdown-label">Registered Credits</span>
              <span class="breakdown-val" id="dispTotalCredits">0.0</span>
            </div>
            <div class="breakdown-item">
```

```
              <span class="breakdown-label">Weighted Points</span>
              <span class="breakdown-val" id="dispWeightedPoints">0.0</span>
            </div>
          </div>
        </div>

        <!-- Simulator Card -->
        <div class="card gpa-simulator-card">
          <div class="metric-top">
            <span class="card-caption">Target SGPA Simulator</span>
            <span class="status-tag tag-neutral">Forecasting</span>
          </div>
          <p class="simulator-intro">Estimate required grade points across remaining semester modules to reach your
target.</p>
          <div class="sim-controls">
            <div class="input-inline">
              <label for="simTargetInput">Target CGPA</label>
              <input type="number" id="simTargetInput" step="0.05" min="5.0" max="10.0" value="9.00">
            </div>
            <button class="btn btn-secondary" id="btnSimulate">Calculate Requirement</button>
          </div>
          <div class="sim-result-box" id="simResultBox">
            Enter target grade above to view credit requirement analysis.
          </div>
        </div>
      </div>

      <!-- Course Table -->
      <div class="card table-card">
        <div class="table-scroll-container">
          <table class="data-table" id="courseTable">
            <thead>
              <tr>
                <th style="min-width: 120px;">Course Code</th>
                <th style="min-width: 180px;">Course Title</th>
                <th style="min-width: 80px; text-align: right;">Credits</th>
                <th style="min-width: 80px; text-align: center;">Grade</th>
                <th style="min-width: 90px; text-align: right;">Points</th>
                <th style="min-width: 110px; text-align: right;">Weighted Pts</th>
                <th style="min-width: 80px; text-align: right;">Action</th>
              </tr>
            </thead>
            <tbody id="courseTableBody">
              <!-- Dynamically populated -->
            </tbody>
          </table>
```

```
          </div>
        </div>
      </section>

      <!-- ==================== TAB 3: POMODORO TIMER ==================== -->
      <section id="tab-pomodoro" class="tab-pane">
        <div class="section-header">
          <div>
            <h1 class="section-title">Focus Timer</h1>
            <p class="section-desc">Structured Study Intervals To Sustain Deliberate Attention.</p>
          </div>
        </div>

        <div class="pomodoro-grid">
          <!-- Main Timer Dial Card -->
          <div class="card timer-card">
            <!-- Mode Selector Presets -->
            <div class="mode-selector">
              <button class="mode-btn active" data-mode="focus" data-time="25">Focus (25m)</button>
              <button class="mode-btn" data-mode="shortBreak" data-time="5">Short Break (5m)</button>
              <button class="mode-btn" data-mode="longBreak" data-time="15">Long Break (15m)</button>
            </div>

            <!-- Editable Duration Strip -->
            <div class="duration-editor-strip">
              <span class="duration-editor-label">Edit Timer Duration:</span>
              <div class="duration-controls-row">
                <input type="number" id="customTimerMinutes" min="1" max="180" value="25" class="duration-input" aria
-label="Timer minutes">
                <span class="duration-unit">min</span>
                <button type="button" class="btn btn-sm btn-secondary" id="btnApplyDuration">Set Duration</button>
              </div>
            </div>

            <div class="dial-container">
              <svg class="dial-svg" width="240" height="240" viewBox="0 0 240 240">
                <circle class="dial-track" cx="120" cy="120" r="104" />
                <circle class="dial-indicator" id="timerProgress" cx="120" cy="120" r="104" />
              </svg>
              <div class="dial-center">
                <span class="timer-readout" id="timerDisplay">25:00</span>
                <span class="timer-status-text" id="timerModeLabel">Focus Interval</span>
              </div>
            </div>

            <div class="timer-action-row">
```

```
            <button class="btn btn-primary btn-md" id="btnTimerStart">
              <span>Start</span>
            </button>
            <button class="btn btn-secondary btn-md" id="btnTimerPause" disabled>
              <span>Pause</span>
            </button>
            <button class="btn btn-secondary btn-md" id="btnTimerReset">
              <span>Reset</span>
            </button>
          </div>

          <div class="timer-meta-strip">
            <div class="meta-stat">
              <span class="meta-num" id="pomoCount">0</span>
              <span class="meta-lbl">Completed Blocks</span>
            </div>
            <div class="meta-stat-divider"></div>
            <div class="meta-stat">
              <span class="meta-num" id="pomoMinutes">0m</span>
              <span class="meta-lbl">Total Focus</span>
            </div>
            <div class="meta-stat-divider"></div>
            <div class="meta-stat">
              <span class="meta-num" id="pomoStreak">0</span>
              <span class="meta-lbl">Session Streak</span>
            </div>
          </div>
        </div>

        <!-- Guidelines & Subject Tagger -->
        <div class="timer-side-column">
          <!-- Subject Selector & Custom Subject Addition -->
          <div class="card">
            <h2 class="card-title">Active Course Focus</h2>

            <div class="form-group">
              <label for="pomoSubjectSelector">Select Subject For Focus Session</label>
              <select id="pomoSubjectSelector" class="form-select">
                <option value="General Focus">General Focus</option>
              </select>
            </div>

            <div class="add-subject-panel">
              <label for="pomoNewSubjectInput">Add New Subject</label>
              <div class="add-subject-input-row">
```

```
                  <input type="text" id="pomoNewSubjectInput" placeholder="e.g. Operating Systems, Machine Learning"
class="form-input">
                  <button type="button" class="btn btn-secondary btn-sm" id="btnAddPomoSubject">+ Add Subject</button
>
                </div>
                <p class="field-hint">Added subjects are saved to your workspace and immediately synchronized with th
e SGPA Calculator.</p>
              </div>
            </div>

            <!-- Protocol Card -->
            <div class="card">
              <h2 class="card-title">Study Session Protocol</h2>
              <ul class="guideline-list">
                <li>
                  <span class="step-num">1</span>
                  <div>
                    <strong>Select One Deliverable</strong>
                    <p>Silence notifications and work only on the selected subject.</p>
                  </div>
                </li>
                <li>
                  <span class="step-num">2</span>
                  <div>
                    <strong>Maintain Continuous Focus</strong>
                    <p>Avoid tab switching or passive browsing until the timer ends.</p>
                  </div>
                </li>
                <li>
                  <span class="step-num">3</span>
                  <div>
                    <strong>Take Deliberate Rest</strong>
                    <p>Step away from the screen for five minutes to recover attention.</p>
                  </div>
                </li>
              </ul>
            </div>
          </div>
        </div>
      </section>

      <!-- ==================== TAB 4: FLASHCARDS ==================== -->
      <section id="tab-flashcards" class="tab-pane">
        <div class="section-header">
          <div>
            <h1 class="section-title">Revision Flashcards</h1>
            <p class="section-desc">Active Recall Practice To Strengthen Concept Retrieval Before Examinations.</p>
          </div>
```

```
          <button class="btn btn-primary" id="btnOpenFlashcardModal">
            <span>New Card</span>
          </button>
        </div>

        <div class="flashcard-deck-layout">
          <div class="flashcard-viewport" id="activeFlashcard">
            <div class="flashcard-box" id="cardInner">
              <!-- Front Face (Question) -->
              <div class="card-side side-front">
                <div class="side-header">
                  <span class="tag-category" id="fcSubject">Active Recall</span>
                  <span class="side-counter" id="fcCounter">0 Cards</span>
                </div>
                <div class="side-content">
                  <span class="content-eyebrow">Prompt</span>
                  <p class="content-text" id="fcQuestion">No revision flashcards added yet. Click '+ New Card' to add
concept prompts for active recall.</p>
                </div>
                <div class="side-footer">
                  <span>Press <kbd class="keycap">Space</kbd> or click to reveal answer</span>
                </div>
              </div>

              <!-- Back Face (Answer) -->
              <div class="card-side side-back">
                <div class="side-header">
                  <span class="tag-category">Answer Explanation</span>
                  <span class="mastery-indicator" id="fcMasteryTag">Unmastered</span>
                </div>
                <div class="side-content">
                  <span class="content-eyebrow">Answer</span>
                  <p class="content-text" id="fcAnswer">Answers will appear here once cards are created.</p>
                </div>
                <div class="side-footer">
                  <span>Click to flip back to question</span>
                </div>
              </div>
            </div>
          </div>

          <!-- Controls -->
          <div class="flashcard-nav-row">
            <button class="btn btn-secondary" id="btnFcPrev">Previous</button>
            <button class="btn btn-secondary" id="btnFcFlip">Flip Card</button>
            <button class="btn btn-secondary" id="btnFcMaster">
```

```
            <span id="fcMasterBtnText">Mark Mastered</span>
          </button>
          <button class="btn btn-secondary" id="btnFcNext">Next</button>
        </div>
      </div>
    </section>

    <!-- ==================== TAB 5: ANALYTICS & EXPORT ==================== -->
    <section id="tab-analytics" class="tab-pane">
      <div class="section-header">
        <div>
          <h1 class="section-title">Academic Report And Export</h1>
          <p class="section-desc">Generate Official Performance Documentation Or Download Local State Backups.</p>
        </div>
        <div class="section-actions">
          <button class="btn btn-secondary" id="btnExportTxt">
            <span>Download Text Report</span>
          </button>
          <button class="btn btn-primary" id="btnExportJson">
            <span>Download JSON Backup</span>
          </button>
        </div>
      </div>

      <div class="analytics-stat-row">
        <div class="card stat-card">
          <span class="stat-caption">Registered Courses</span>
          <span class="stat-metric-large" id="statCourseCount">0</span>
        </div>
        <div class="card stat-card">
          <span class="stat-caption">Logged Assignments</span>
          <span class="stat-metric-large" id="statTaskCount">0</span>
        </div>
        <div class="card stat-card">
          <span class="stat-caption">Mastered Concepts</span>
          <span class="stat-metric-large" id="statMasteredCount">0 / 0</span>
        </div>
        <div class="card stat-card">
          <span class="stat-caption">Total Focus Logged</span>
          <span class="stat-metric-large" id="statStudyMins">0m</span>
        </div>
      </div>

      <div class="card report-box">
        <div class="report-box-header">
```

```
            <span class="card-title">Generated Summary Preview</span>
            <span class="status-tag tag-neutral">ASCII Formatted</span>
          </div>
          <pre class="ascii-preview" id="reportPreviewBox">Loading academic summary...</pre>
        </div>
      </section>

    </main>

  </div>

  <!-- Shared Datalist for Autocomplete Subjects -->
  <datalist id="subjectListDatalist"></datalist>

  <!-- Modal 1: Assignment -->
  <div class="modal-backdrop" id="taskModal">
    <div class="modal-dialog">
      <div class="modal-header">
        <h2 class="modal-title">New Assignment</h2>
        <button class="btn-close" id="btnCloseTaskModal">X</button>
      </div>
      <form id="taskForm">
        <div class="form-group">
          <label for="taskTitleInput">Assignment Title</label>
          <input type="text" id="taskTitleInput" required placeholder="e.g. Multithreaded Producer-Consumer Practical
">
        </div>
        <div class="form-row-2">
          <div class="form-group">
            <label for="taskSubjectInput">Course Or Module</label>
            <input type="text" id="taskSubjectInput" list="subjectListDatalist" required placeholder="e.g. Object Ori
ented Programming">
          </div>
          <div class="form-group">
            <label for="taskDateInput">Due Date</label>
            <input type="date" id="taskDateInput" required>
          </div>
        </div>
        <div class="form-row-2">
          <div class="form-group">
            <label for="taskPriorityInput">Priority</label>
            <select id="taskPriorityInput">
              <option value="High">High</option>
              <option value="Medium" selected>Medium</option>
              <option value="Low">Low</option>
            </select>
          </div>
```

```
          <div class="form-group">
            <label for="taskHoursInput">Estimated Hours</label>
            <input type="number" id="taskHoursInput" min="1" max="50" value="3">
          </div>
        </div>
        <div class="modal-actions">
          <button type="button" class="btn btn-secondary" id="btnCancelTask">Cancel</button>
          <button type="submit" class="btn btn-primary">Save Assignment</button>
        </div>
      </form>
    </div>
  </div>

  <!-- Modal 2: Course -->
  <div class="modal-backdrop" id="courseModal">
    <div class="modal-dialog">
      <div class="modal-header">
        <h2 class="modal-title">Add Semester Course</h2>
        <button class="btn-close" id="btnCloseCourseModal">X</button>
      </div>
      <form id="courseForm">
        <div class="form-row-2">
          <div class="form-group">
            <label for="courseCodeInput">Course Code</label>
            <input type="text" id="courseCodeInput" required placeholder="e.g. 25CSH-208">
          </div>
          <div class="form-group">
            <label for="courseNameInput">Course Title</label>
            <input type="text" id="courseNameInput" list="subjectListDatalist" required placeholder="e.g. Object Orie
nted Programming">
          </div>
        </div>
        <div class="form-row-2">
          <div class="form-group">
            <label for="courseCreditsInput">Credits</label>
            <select id="courseCreditsInput">
              <option value="1.0">1.0</option>
              <option value="2.0">2.0</option>
              <option value="3.0">3.0</option>
              <option value="3.5">3.5</option>
              <option value="4.0" selected>4.0</option>
              <option value="5.0">5.0</option>
            </select>
          </div>
          <div class="form-group">
            <label for="courseGradeInput">Expected Grade</label>
```

```
            <select id="courseGradeInput">
              <option value="A+">A+ (10.0)</option>
              <option value="A" selected>A (9.0)</option>
              <option value="B+">B+ (8.0)</option>
              <option value="B">B (7.0)</option>
              <option value="C+">C+ (6.0)</option>
              <option value="C">C (5.0)</option>
              <option value="D">D (4.0)</option>
              <option value="F">F (0.0)</option>
            </select>
          </div>
        </div>
        <div class="modal-actions">
          <button type="button" class="btn btn-secondary" id="btnCancelCourse">Cancel</button>
          <button type="submit" class="btn btn-primary">Add Course</button>
        </div>
      </form>
    </div>
  </div>

  <!-- Modal 3: Flashcard -->
  <div class="modal-backdrop" id="flashcardModal">
    <div class="modal-dialog">
      <div class="modal-header">
        <h2 class="modal-title">Create Revision Card</h2>
        <button class="btn-close" id="btnCloseFcModal">X</button>
      </div>
      <form id="flashcardForm">
        <div class="form-group">
          <label for="fcSubjectInput">Topic Or Subject</label>
          <input type="text" id="fcSubjectInput" list="subjectListDatalist" required placeholder="e.g. Java Concurren
cy, Data Structures">
        </div>
        <div class="form-group">
          <label for="fcQuestionInput">Question Prompt</label>
          <textarea id="fcQuestionInput" rows="3" required placeholder="Enter concept prompt or question..."></textar
ea>
        </div>
        <div class="form-group">
          <label for="fcAnswerInput">Answer Explanation</label>
          <textarea id="fcAnswerInput" rows="4" required placeholder="Enter concise answer explanation..."></textarea
>
        </div>
        <div class="modal-actions">
          <button type="button" class="btn btn-secondary" id="btnCancelFc">Cancel</button>
          <button type="submit" class="btn btn-primary">Save Card</button>
        </div>
      </form>
    </div>
  </div>

  <script src="app.js"></script>
</body>
</html>
```

## 2. Client Logic: app.js (WebAssembly SQLite Engine & Reactive Controllers)

```
/**
 * Student Workspace • Application Core Engine
 * Client-Side Offline Architecture powered by WebAssembly SQLite (sql.js)
 * Fully compatible with GitHub Pages & Vercel Static Hosting
 */

const GRADE_POINTS = {
  "A+": 10.0,
  "A": 9.0,
  "B+": 8.0,
  "B": 7.0,
  "C+": 6.0,
  "C": 5.0,
  "D": 4.0,
  "F": 0.0
};

// ============================================================================
// SQLite Database Layer (WebAssembly & Local Persistence)
// ============================================================================
class DatabaseManager {
  static STORAGE_KEY = "student_workspace_sqlite_binary";
  static FALLBACK_KEY = "student_workspace_fallback_json";

  constructor() {
    this.db = null;
    this.isSqlJsReady = false;
    // In-memory data cache
    this.cache = {
      courses: [],
      tasks: [],
      flashcards: [],
      customSubjects: [],
      pomodoroStats: {
        completedSessions: 0,
        totalMinutes: 0,
        streakDays: 0
      }
    };
  }

  async init() {
    try {
      if (typeof window.initSqlJs === "function") {
        const SQL = await window.initSqlJs({
```

```
        locateFile: file => `https://cdnjs.cloudflare.com/ajax/libs/sql.js/1.8.0/${file}`
      });

      const saved = localStorage.getItem(DatabaseManager.STORAGE_KEY);
      if (saved) {
        try {
          const u8 = new Uint8Array(JSON.parse(saved));
          this.db = new SQL.Database(u8);
        } catch (e) {
          console.warn("Could not parse saved SQLite binary, creating fresh DB:", e);
          this.db = new SQL.Database();
        }
      } else {
        this.db = new SQL.Database();
      }

      this.initSchema();
      this.isSqlJsReady = true;
      this.loadFromSql();
      return;
    }
  } catch (err) {
    console.warn("WebAssembly SQLite initialization notice (using fallback engine):", err);
  }

  // Fallback: LocalStorage JSON storage
  this.loadFromFallback();
}

initSchema() {
  if (!this.db) return;
  this.db.run(`
    CREATE TABLE IF NOT EXISTS courses (
      id TEXT PRIMARY KEY,
      code TEXT NOT NULL,
      name TEXT NOT NULL,
      credits REAL NOT NULL,
      grade TEXT NOT NULL,
      grade_point REAL NOT NULL,
      created_at INTEGER NOT NULL
    );
    CREATE TABLE IF NOT EXISTS tasks (
      id TEXT PRIMARY KEY,
      title TEXT NOT NULL,
      course TEXT NOT NULL,
```

```
        due_date TEXT NOT NULL,
        priority TEXT NOT NULL,
        hours INTEGER NOT NULL,
        completed INTEGER NOT NULL DEFAULT 0,
        created_at INTEGER NOT NULL
      );
      CREATE TABLE IF NOT EXISTS flashcards (
        id TEXT PRIMARY KEY,
        subject TEXT NOT NULL,
        question TEXT NOT NULL,
        answer TEXT NOT NULL,
        mastered INTEGER NOT NULL DEFAULT 0,
        created_at INTEGER NOT NULL
      );
      CREATE TABLE IF NOT EXISTS pomodoro_sessions (
        id TEXT PRIMARY KEY,
        subject TEXT NOT NULL,
        duration_minutes INTEGER NOT NULL,
        completed_at INTEGER NOT NULL
      );
      CREATE TABLE IF NOT EXISTS custom_subjects (
        name TEXT PRIMARY KEY,
        created_at INTEGER NOT NULL
      );
    `);
    // Note: Database starts unseeded per specification.
  }

  persist() {
    if (this.isSqlJsReady && this.db) {
      try {
        const binaryArray = this.db.export();
        localStorage.setItem(DatabaseManager.STORAGE_KEY, JSON.stringify(Array.from(binaryArray)));
      } catch (e) {
        console.warn("Storage quota warning on persisting SQLite DB:", e);
      }
    } else {
      try {
        localStorage.setItem(DatabaseManager.FALLBACK_KEY, JSON.stringify(this.cache));
      } catch (e) {
        console.warn("Storage quota warning on fallback save:", e);
      }
    }
  }
```

```
  loadFromSql() {
    if (!this.db) return;

    // Load courses
    const courseStmt = this.db.prepare("SELECT * FROM courses ORDER BY created_at ASC");
    const courses = [];
    while (courseStmt.step()) {
      const row = courseStmt.getAsObject();
      courses.push({
        id: row.id,
        code: row.code,
        name: row.name,
        credits: parseFloat(row.credits),
        grade: row.grade,
        gradePoint: parseFloat(row.grade_point)
      });
    }
    courseStmt.free();
    this.cache.courses = courses;

    // Load custom subjects
    try {
      const subStmt = this.db.prepare("SELECT name FROM custom_subjects ORDER BY created_at ASC");
      const subs = [];
      while (subStmt.step()) {
        const row = subStmt.getAsObject();
        if (row.name) subs.push(row.name);
      }
      subStmt.free();
      this.cache.customSubjects = subs;
    } catch (e) {
      this.cache.customSubjects = [];
    }

    // Load tasks
    const taskStmt = this.db.prepare("SELECT * FROM tasks ORDER BY created_at DESC");
    const tasks = [];
    while (taskStmt.step()) {
      const row = taskStmt.getAsObject();
      tasks.push({
        id: row.id,
        title: row.title,
        course: row.course,
        dueDate: row.due_date,
        priority: row.priority,
```

```
        hours: parseInt(row.hours, 10),
        completed: Boolean(row.completed)
      });
    }
    taskStmt.free();
    this.cache.tasks = tasks;

    // Load flashcards
    const fcStmt = this.db.prepare("SELECT * FROM flashcards ORDER BY created_at ASC");
    const flashcards = [];
    while (fcStmt.step()) {
      const row = fcStmt.getAsObject();
      flashcards.push({
        id: row.id,
        subject: row.subject,
        question: row.question,
        answer: row.answer,
        mastered: Boolean(row.mastered)
      });
    }
    fcStmt.free();
    this.cache.flashcards = flashcards;

    // Load pomodoro stats
    const pomoRes = this.db.exec("SELECT COUNT(*) as count, COALESCE(SUM(duration_minutes), 0) as total FROM pomodoro
_sessions");
    let count = 0;
    let total = 0;
    if (pomoRes.length > 0 && pomoRes[0].values.length > 0) {
      count = pomoRes[0].values[0][0] || 0;
      total = pomoRes[0].values[0][1] || 0;
    }
    this.cache.pomodoroStats = {
      completedSessions: count,
      totalMinutes: total,
      streakDays: count > 0 ? Math.min(count, 4) : 0
    };
  }

  loadFromFallback() {
    try {
      const raw = localStorage.getItem(DatabaseManager.FALLBACK_KEY);
      if (raw) {
        this.cache = JSON.parse(raw);
        if (!this.cache.customSubjects) this.cache.customSubjects = [];
        return;
```

```
      }
    } catch (e) {}

    // Clean initial state (unseeded)
    this.cache = {
      courses: [],
      tasks: [],
      flashcards: [],
      customSubjects: [],
      pomodoroStats: { completedSessions: 0, totalMinutes: 0, streakDays: 0 }
    };
  }

  // --- Unified Subject Management ---
  getAllSubjects() {
    const fromCourses = (this.cache.courses || []).map(c => c.name ? c.name.trim() : "").filter(Boolean);
    const fromCustom = (this.cache.customSubjects || []).map(s => s ? s.trim() : "").filter(Boolean);
    const combined = Array.from(new Set([...fromCourses, ...fromCustom]));
    return combined;
  }

  addSubject(subjectName) {
    const trimmed = (subjectName || "").trim();
    if (!trimmed) return;

    if (!this.cache.customSubjects) this.cache.customSubjects = [];
    if (!this.cache.customSubjects.includes(trimmed)) {
      this.cache.customSubjects.push(trimmed);
      if (this.isSqlJsReady && this.db) {
        try {
          this.db.run("INSERT OR IGNORE INTO custom_subjects (name, created_at) VALUES (?, ?)", [trimmed, Date.now()]
);
        } catch (e) {}
      }
    }

    // Also register subject into courses table if not already present
    const existing = this.cache.courses.find(c => c.name.toLowerCase() === trimmed.toLowerCase());
    if (!existing) {
      const codeIndex = this.cache.courses.length + 1;
      const code = "SUB-" + String(codeIndex).padStart(2, '0');
      this.addCourse({
        id: "C-" + Date.now(),
        code: code,
        name: trimmed,
        credits: 4.0,
```

```
        grade: "A"
      });
    } else {
      this.persist();
    }
  }

  // --- Task Operations ---
  addTask(task) {
    this.cache.tasks.unshift(task);
    if (this.isSqlJsReady && this.db) {
      this.db.run(
        "INSERT INTO tasks (id, title, course, due_date, priority, hours, completed, created_at) VALUES (?, ?, ?, ?,
?, ?, ?, ?)",
        [task.id, task.title, task.course, task.dueDate, task.priority, task.hours, task.completed ? 1 : 0, Date.now(
)]
      );
    }
    this.persist();
  }

  toggleTask(id) {
    const t = this.cache.tasks.find(x => x.id === id);
    if (t) {
      t.completed = !t.completed;
      if (this.isSqlJsReady && this.db) {
        this.db.run("UPDATE tasks SET completed = ? WHERE id = ?", [t.completed ? 1 : 0, id]);
      }
      this.persist();
    }
  }

  deleteTask(id) {
    this.cache.tasks = this.cache.tasks.filter(x => x.id !== id);
    if (this.isSqlJsReady && this.db) {
      this.db.run("DELETE FROM tasks WHERE id = ?", [id]);
    }
    this.persist();
  }

  // --- Course Operations ---
  addCourse(course) {
    const gradePoint = GRADE_POINTS[course.grade] ?? 0.0;
    course.gradePoint = gradePoint;
    this.cache.courses.push(course);
    if (this.isSqlJsReady && this.db) {
      this.db.run(
```

```
      "INSERT INTO courses (id, code, name, credits, grade, grade_point, created_at) VALUES (?, ?, ?, ?, ?, ?, ?)",
      [course.id, course.code, course.name, course.credits, course.grade, gradePoint, Date.now()]
    );
  }
  this.persist();
}

deleteCourse(id) {
  this.cache.courses = this.cache.courses.filter(x => x.id !== id);
  if (this.isSqlJsReady && this.db) {
    this.db.run("DELETE FROM courses WHERE id = ?", [id]);
  }
  this.persist();
}

// --- Flashcard Operations ---
addFlashcard(card) {
  this.cache.flashcards.push(card);
  if (this.isSqlJsReady && this.db) {
    this.db.run(
      "INSERT INTO flashcards (id, subject, question, answer, mastered, created_at) VALUES (?, ?, ?, ?, ?, ?)",
      [card.id, card.subject, card.question, card.answer, card.mastered ? 1 : 0, Date.now()]
    );
  }
  this.persist();
}

toggleFlashcardMastery(id) {
  const fc = this.cache.flashcards.find(x => x.id === id);
  if (fc) {
    fc.mastered = !fc.mastered;
    if (this.isSqlJsReady && this.db) {
      this.db.run("UPDATE flashcards SET mastered = ? WHERE id = ?", [fc.mastered ? 1 : 0, id]);
    }
    this.persist();
  }
}

// --- Pomodoro Operations ---
logPomodoroSession(subject, duration) {
  this.cache.pomodoroStats.completedSessions++;
  this.cache.pomodoroStats.totalMinutes += duration;
  this.cache.pomodoroStats.streakDays = Math.max(1, this.cache.pomodoroStats.streakDays);

  if (this.isSqlJsReady && this.db) {
```

```
      const id = "PS-" + Date.now();
      this.db.run(
        "INSERT INTO pomodoro_sessions (id, subject, duration_minutes, completed_at) VALUES (?, ?, ?, ?)",
        [id, subject, duration, Date.now()]
      );
    }
    this.persist();
  }
}

// App Singleton Instance
const dbMgr = new DatabaseManager();

// ============================================================================
// Initialization & Navigation
// ============================================================================
document.addEventListener("DOMContentLoaded", async () => {
  await dbMgr.init();
  initTabs();
  initAssignments();
  initGpaCalculator();
  initPomodoro();
  initFlashcards();
  initAnalytics();
  updateSubjectSelectors();
  updateHeaderBadges();
});

function initTabs() {
  const tabs = document.querySelectorAll(".nav-tab");
  tabs.forEach(tab => {
    tab.addEventListener("click", () => {
      switchTab(tab.getAttribute("data-tab"));
    });
  });

  const hash = window.location.hash.replace("#", "");
  if (hash && ["assignments", "gpa", "pomodoro", "flashcards", "analytics"].includes(hash)) {
    switchTab(hash);
  }
}

function switchTab(tabName) {
  const tabs = document.querySelectorAll(".nav-tab");
  tabs.forEach(t => {
```

```
    t.classList.toggle("active", t.getAttribute("data-tab") === tabName);
  });

  const targetPaneId = "tab-" + tabName;
  document.querySelectorAll(".tab-pane").forEach(pane => {
    pane.classList.toggle("active", pane.id === targetPaneId);
  });

  if (tabName === "analytics") {
    renderAnalytics();
  } else if (tabName === "pomodoro" || tabName === "gpa") {
    updateSubjectSelectors();
  }
}

function updateHeaderBadges() {
  const quickSgpa = document.getElementById("quickSgpa");
  const quickPending = document.getElementById("quickPending");
  const quickPomodoro = document.getElementById("quickPomodoro");

  const sgpa = calculateSgpa().sgpa;
  if (quickSgpa) quickSgpa.textContent = sgpa > 0 ? sgpa.toFixed(2) : "0.00";

  const pendingCount = dbMgr.cache.tasks.filter(t => !t.completed).length;
  if (quickPending) quickPending.textContent = pendingCount;

  if (quickPomodoro) quickPomodoro.textContent = dbMgr.cache.pomodoroStats.completedSessions;
}

// ============================================================================
// Subject Selectors & Autocomplete Synchronization
// ============================================================================
function updateSubjectSelectors(selectedSubject = null) {
  const subjects = dbMgr.getAllSubjects();

  // 1. Update Pomodoro Selector
  const pomoSelector = document.getElementById("pomoSubjectSelector");
  if (pomoSelector) {
    const prevValue = selectedSubject || pomoSelector.value;
    pomoSelector.innerHTML = "";

    if (subjects.length === 0) {
      const opt = document.createElement("option");
      opt.value = "General Focus";
      opt.textContent = "General Focus (Default)";
```

```
      pomoSelector.appendChild(opt);
    } else {
      subjects.forEach(sub => {
        const opt = document.createElement("option");
        opt.value = sub;
        opt.textContent = sub;
        pomoSelector.appendChild(opt);
      });
    }

    if (prevValue && subjects.includes(prevValue)) {
      pomoSelector.value = prevValue;
    }
  }

  // 2. Update Shared Datalist for Autocomplete
  const datalist = document.getElementById("subjectListDatalist");
  if (datalist) {
    datalist.innerHTML = subjects.map(s => `<option value="${escapeHtml(s)}">`).join("");
  }
}

// ============================================================================
// Module 1: Assignments & Tasks
// ============================================================================
function initAssignments() {
  const searchInput = document.getElementById("taskSearchInput");
  const priorityFilter = document.getElementById("priorityFilter");
  const statusFilter = document.getElementById("statusFilter");
  const btnSortUrgency = document.getElementById("btnSortUrgency");
  const btnOpenModal = document.getElementById("btnOpenTaskModal");
  const taskModal = document.getElementById("taskModal");
  const btnCloseModal = document.getElementById("btnCloseTaskModal");
  const btnCancelTask = document.getElementById("btnCancelTask");
  const taskForm = document.getElementById("taskForm");

  renderTasks();

  if (searchInput) searchInput.addEventListener("input", renderTasks);
  if (priorityFilter) priorityFilter.addEventListener("change", renderTasks);
  if (statusFilter) statusFilter.addEventListener("change", renderTasks);

  if (btnSortUrgency) {
    btnSortUrgency.addEventListener("click", () => {
      dbMgr.cache.tasks.sort((a, b) => {
```

```
      const pOrder = { "High": 1, "Medium": 2, "Low": 3 };
      if (pOrder[a.priority] !== pOrder[b.priority]) {
        return pOrder[a.priority] - pOrder[b.priority];
      }
      return new Date(a.dueDate) - new Date(b.dueDate);
    });
    renderTasks();
  });
}

if (btnOpenModal) {
  btnOpenModal.addEventListener("click", () => {
    updateSubjectSelectors();
    taskModal.classList.add("active");
  });
}
const closeModal = () => taskModal.classList.remove("active");
if (btnCloseModal) btnCloseModal.addEventListener("click", closeModal);
if (btnCancelTask) btnCancelTask.addEventListener("click", closeModal);

if (taskForm) {
  taskForm.addEventListener("submit", (e) => {
    e.preventDefault();
    const courseSubject = document.getElementById("taskSubjectInput").value.trim();
    const newTask = {
      id: "T-" + (Date.now() % 100000),
      title: document.getElementById("taskTitleInput").value.trim(),
      course: courseSubject,
      dueDate: document.getElementById("taskDateInput").value,
      priority: document.getElementById("taskPriorityInput").value,
      hours: parseInt(document.getElementById("taskHoursInput").value, 10) || 3,
      completed: false
    };
    dbMgr.addTask(newTask);

    // Register subject into workspace subjects
    if (courseSubject) {
      dbMgr.addSubject(courseSubject);
      updateSubjectSelectors();
    }

    renderTasks();
    updateHeaderBadges();
    taskForm.reset();
    closeModal();
```

```
    });
  }
}

function renderTasks() {
  const taskList = document.getElementById("taskList");
  if (!taskList) return;

  const searchQuery = (document.getElementById("taskSearchInput")?.value || "").toLowerCase();
  const priorityVal = document.getElementById("priorityFilter")?.value || "ALL";
  const statusVal = document.getElementById("statusFilter")?.value || "ALL";

  const filtered = dbMgr.cache.tasks.filter(t => {
    const matchesSearch = t.title.toLowerCase().includes(searchQuery) || t.course.toLowerCase().includes(searchQuery)
;
    const matchesPriority = (priorityVal === "ALL" || t.priority === priorityVal);
    const matchesStatus = (statusVal === "ALL" || (statusVal === "COMPLETED" ? t.completed : !t.completed));
    return matchesSearch && matchesPriority && matchesStatus;
  });

  if (filtered.length === 0) {
    taskList.innerHTML = `
      <div class="empty-state-card">
        <p>No assignments recorded yet.</p>
        <div class="empty-state-action">
          <button class="btn btn-sm btn-primary" onclick="document.getElementById('taskModal').classList.add('active'
)">
            + New Assignment
          </button>
        </div>
      </div>
    `;
    return;
  }

  taskList.innerHTML = filtered.map(t => `
    <div class="task-card ${t.completed ? 'completed' : ''}" data-id="${t.id}">
      <div class="task-card-header">
        <div>
          <h2 class="task-title">${escapeHtml(t.title)}</h2>
          <p class="task-course">${escapeHtml(t.course)}</p>
        </div>
        <span class="priority-tag priority-${t.priority.toLowerCase()}">${t.priority}</span>
      </div>
      <div class="task-meta-row">
        <span>Due ${t.dueDate}</span>
        <span>${t.hours} hrs est.</span>
```

```
      </div>
      <div class="task-footer-actions">
        <button class="btn btn-sm btn-secondary" onclick="toggleTaskComplete('${t.id}')">
          ${t.completed ? 'Reopen' : 'Mark Completed'}
        </button>
        <button class="btn btn-sm btn-ghost" onclick="deleteTaskItem('${t.id}')">
          Delete
        </button>
      </div>
    </div>
  `).join("");
}

window.toggleTaskComplete = function(id) {
  dbMgr.toggleTask(id);
  renderTasks();
  updateHeaderBadges();
};

window.deleteTaskItem = function(id) {
  dbMgr.deleteTask(id);
  renderTasks();
  updateHeaderBadges();
};

// ============================================================================
// Module 2: SGPA Grade Calculator
// ============================================================================
function calculateSgpa() {
  let totalCredits = 0;
  let totalWeighted = 0;
  dbMgr.cache.courses.forEach(c => {
    const pt = GRADE_POINTS[c.grade] ?? c.gradePoint ?? 0.0;
    totalCredits += c.credits;
    totalWeighted += (c.credits * pt);
  });
  const sgpa = totalCredits > 0 ? (totalWeighted / totalCredits) : 0.0;
  return { sgpa, totalCredits, totalWeighted };
}

function initGpaCalculator() {
  const btnOpenModal = document.getElementById("btnOpenCourseModal");
  const courseModal = document.getElementById("courseModal");
  const btnCloseModal = document.getElementById("btnCloseCourseModal");
  const btnCancelCourse = document.getElementById("btnCancelCourse");
```

```
  const courseForm = document.getElementById("courseForm");
  const btnSimulate = document.getElementById("btnSimulate");

  renderCourseTable();

  if (btnOpenModal) {
    btnOpenModal.addEventListener("click", () => {
      updateSubjectSelectors();
      courseModal.classList.add("active");
    });
  }
  const closeModal = () => courseModal.classList.remove("active");
  if (btnCloseModal) btnCloseModal.addEventListener("click", closeModal);
  if (btnCancelCourse) btnCancelCourse.addEventListener("click", closeModal);

  if (courseForm) {
    courseForm.addEventListener("submit", (e) => {
      e.preventDefault();
      const courseName = document.getElementById("courseNameInput").value.trim();
      const newCourse = {
        id: "C-" + (Date.now() % 100000),
        code: document.getElementById("courseCodeInput").value.trim(),
        name: courseName,
        credits: parseFloat(document.getElementById("courseCreditsInput").value),
        grade: document.getElementById("courseGradeInput").value
      };
      dbMgr.addCourse(newCourse);
      dbMgr.addSubject(courseName);

      renderCourseTable();
      updateSubjectSelectors();
      updateHeaderBadges();
      courseForm.reset();
      closeModal();
    });
  }

  if (btnSimulate) {
    btnSimulate.addEventListener("click", runTargetSimulation);
  }
}

function renderCourseTable() {
  const tbody = document.getElementById("courseTableBody");
  if (!tbody) return;
```

```
const { sgpa, totalCredits, totalWeighted } = calculateSgpa();

const dispSgpa = document.getElementById("dispSgpa");
const dispClassification = document.getElementById("dispClassification");
const dispTotalCredits = document.getElementById("dispTotalCredits");
const dispWeightedPoints = document.getElementById("dispWeightedPoints");

if (dispSgpa) dispSgpa.textContent = sgpa > 0 ? sgpa.toFixed(2) : "0.00";
if (dispTotalCredits) dispTotalCredits.textContent = totalCredits.toFixed(1);
if (dispWeightedPoints) dispWeightedPoints.textContent = totalWeighted.toFixed(1);

if (dispClassification) {
  if (totalCredits === 0) {
    dispClassification.textContent = "No courses enrolled";
    dispClassification.className = "status-tag tag-neutral";
  } else if (sgpa >= 9.0) {
    dispClassification.textContent = "First Class With Distinction";
    dispClassification.className = "status-tag tag-green";
  } else if (sgpa >= 8.0) {
    dispClassification.textContent = "First Class With Honors";
    dispClassification.className = "status-tag tag-neutral";
  } else if (sgpa >= 6.5) {
    dispClassification.textContent = "First Class";
    dispClassification.className = "status-tag tag-neutral";
  } else {
    dispClassification.textContent = "Passing Standing";
    dispClassification.className = "status-tag tag-neutral";
  }
}

if (dbMgr.cache.courses.length === 0) {
  tbody.innerHTML = `
    <tr>
      <td colspan="7" class="empty-state-cell">
        No courses added yet. Click "+ Add Course" to register modules and calculate your SGPA.
      </td>
    </tr>
  `;
  return;
}

tbody.innerHTML = dbMgr.cache.courses.map(c => {
  const pt = GRADE_POINTS[c.grade] ?? 0.0;
  const weighted = (c.credits * pt).toFixed(1);
```

```
    return `
      <tr>
        <td><span class="code-pill">${escapeHtml(c.code)}</span></td>
        <td>${escapeHtml(c.name)}</td>
        <td style="text-align: right;">${c.credits.toFixed(1)}</td>
        <td style="text-align: center;"><span class="status-tag tag-neutral">${c.grade}</span></td>
        <td style="text-align: right;">${pt.toFixed(1)}</td>
        <td style="text-align: right;"><strong>${weighted}</strong></td>
        <td style="text-align: right;">
          <button class="btn btn-sm btn-ghost" onclick="deleteCourseItem('${c.id}')">
            Remove
          </button>
        </td>
      </tr>
    `;
  }).join("");
}

window.deleteCourseItem = function(id) {
  dbMgr.deleteCourse(id);
  renderCourseTable();
  updateSubjectSelectors();
  updateHeaderBadges();
};

function runTargetSimulation() {
  const target = parseFloat(document.getElementById("simTargetInput")?.value || 0);
  const simResultBox = document.getElementById("simResultBox");
  if (!simResultBox) return;

  const { sgpa, totalCredits, totalWeighted } = calculateSgpa();
  if (totalCredits === 0) {
    simResultBox.textContent = "Add courses first to simulate target grades.";
    return;
  }

  const diff = target - sgpa;
  if (diff <= 0) {
    simResultBox.innerHTML = `Target Achieved: Current SGPA (${sgpa.toFixed(2)}) meets or exceeds your goal (${target
.toFixed(2)}).`;
  } else {
    const requiredTotal = target * (totalCredits + 4);
    const neededOnNext = requiredTotal - totalWeighted;
    const gradeNeeded = (neededOnNext / 4).toFixed(1);
    simResultBox.innerHTML = `To reach <strong>${target.toFixed(2)}</strong>, you need an average grade point of <str
ong>${gradeNeeded} / 10</strong> on the next 4-credit course.`;
  }
```

```
}

// ============================================================================
// Module 3: Pomodoro Focus Timer (Editable Duration & Custom Subjects)
// ============================================================================
let timerInterval = null;
let initialSeconds = 25 * 60;
let timerSeconds = 25 * 60;
let isTimerRunning = false;
let currentTimerMode = "focus";

function initPomodoro() {
  const btnStart = document.getElementById("btnTimerStart");
  const btnPause = document.getElementById("btnTimerPause");
  const btnReset = document.getElementById("btnTimerReset");
  const modeBtns = document.querySelectorAll(".mode-btn");
  const customInput = document.getElementById("customTimerMinutes");
  const btnApplyDuration = document.getElementById("btnApplyDuration");
  const btnAddSubject = document.getElementById("btnAddPomoSubject");
  const newSubjectInput = document.getElementById("pomoNewSubjectInput");

  // Mode preset buttons
  modeBtns.forEach(btn => {
    btn.addEventListener("click", () => {
      modeBtns.forEach(b => b.classList.remove("active"));
      btn.classList.add("active");
      const mode = btn.getAttribute("data-mode");
      const mins = parseInt(btn.getAttribute("data-time"), 10);

      if (customInput) customInput.value = mins;
      setTimerDuration(mode, mins);
    });
  });

  // Custom duration setter
  const applyCustomDuration = () => {
    if (!customInput) return;
    let mins = parseInt(customInput.value, 10);
    if (isNaN(mins) || mins < 1) mins = 1;
    if (mins > 180) mins = 180;
    customInput.value = mins;

    // Remove active from presets if not matching standard preset
    modeBtns.forEach(b => {
      const bMins = parseInt(b.getAttribute("data-time"), 10);
```

```
      b.classList.toggle("active", bMins === mins);
    });

    setTimerDuration("focus", mins);
  };

  if (btnApplyDuration) btnApplyDuration.addEventListener("click", applyCustomDuration);
  if (customInput) {
    customInput.addEventListener("keydown", (e) => {
      if (e.key === "Enter") {
        e.preventDefault();
        applyCustomDuration();
      }
    });
  }

  // Add custom subject button
  const handleAddNewSubject = () => {
    if (!newSubjectInput) return;
    const name = newSubjectInput.value.trim();
    if (!name) return;

    dbMgr.addSubject(name);
    updateSubjectSelectors(name);
    renderCourseTable();
    updateHeaderBadges();
    newSubjectInput.value = "";
  };

  if (btnAddSubject) btnAddSubject.addEventListener("click", handleAddNewSubject);
  if (newSubjectInput) {
    newSubjectInput.addEventListener("keydown", (e) => {
      if (e.key === "Enter") {
        e.preventDefault();
        handleAddNewSubject();
      }
    });
  }

  if (btnStart) btnStart.addEventListener("click", startTimer);
  if (btnPause) btnPause.addEventListener("click", pauseTimer);
  if (btnReset) btnReset.addEventListener("click", resetTimer);

  updateTimerDisplay();
  updatePomodoroStats();
```

```
}

function setTimerDuration(mode, minutes) {
  pauseTimer();
  currentTimerMode = mode;
  initialSeconds = minutes * 60;
  timerSeconds = initialSeconds;

  const modeLabel = document.getElementById("timerModeLabel");
  if (modeLabel) {
    if (mode === "shortBreak") modeLabel.textContent = `Short Break (${minutes}m)`;
    else if (mode === "longBreak") modeLabel.textContent = `Long Break (${minutes}m)`;
    else modeLabel.textContent = `Focus Interval (${minutes}m)`;
  }

  updateTimerDisplay();
}

function startTimer() {
  if (isTimerRunning) return;
  isTimerRunning = true;
  document.getElementById("btnTimerStart").disabled = true;
  document.getElementById("btnTimerPause").disabled = false;

  timerInterval = setInterval(() => {
    if (timerSeconds > 0) {
      timerSeconds--;
      updateTimerDisplay();
    } else {
      clearInterval(timerInterval);
      isTimerRunning = false;
      document.getElementById("btnTimerStart").disabled = false;
      document.getElementById("btnTimerPause").disabled = true;

      if (currentTimerMode === "focus") {
        const subjectSelector = document.getElementById("pomoSubjectSelector");
        const subject = subjectSelector ? subjectSelector.value : "General Focus";
        dbMgr.logPomodoroSession(subject, Math.round(initialSeconds / 60));
        updatePomodoroStats();
        updateHeaderBadges();
      }
      resetTimer();
    }
  }, 1000);
}
```

```
function pauseTimer() {
  if (!isTimerRunning) return;
  clearInterval(timerInterval);
  isTimerRunning = false;
  document.getElementById("btnTimerStart").disabled = false;
  document.getElementById("btnTimerPause").disabled = true;
}

function resetTimer() {
  pauseTimer();
  timerSeconds = initialSeconds;
  updateTimerDisplay();
}

function updateTimerDisplay() {
  const m = Math.floor(timerSeconds / 60);
  const s = timerSeconds % 60;
  const str = `${String(m).padStart(2, '0')}:${String(s).padStart(2, '0')}`;

  const disp = document.getElementById("timerDisplay");
  if (disp) disp.textContent = str;

  const ring = document.getElementById("timerProgress");
  if (ring) {
    const totalCircumference = 2 * Math.PI * 104; // r = 104
    const fraction = initialSeconds > 0 ? (timerSeconds / initialSeconds) : 0;
    const offset = totalCircumference * (1 - fraction);
    ring.style.strokeDashoffset = offset;
  }
}

function updatePomodoroStats() {
  const pomoCount = document.getElementById("pomoCount");
  const pomoMinutes = document.getElementById("pomoMinutes");
  const pomoStreak = document.getElementById("pomoStreak");

  if (pomoCount) pomoCount.textContent = dbMgr.cache.pomodoroStats.completedSessions;
  if (pomoMinutes) pomoMinutes.textContent = `${dbMgr.cache.pomodoroStats.totalMinutes}m`;
  if (pomoStreak) pomoStreak.textContent = dbMgr.cache.pomodoroStats.streakDays;
}

// ============================================================================
// Module 4: Revision Flashcards
// ============================================================================
```

```
let currentCardIndex = 0;

function initFlashcards() {
  const cardInner = document.getElementById("cardInner");
  const btnFlip = document.getElementById("btnFcFlip");
  const btnPrev = document.getElementById("btnFcPrev");
  const btnNext = document.getElementById("btnFcNext");
  const btnMaster = document.getElementById("btnFcMaster");
  const btnOpenModal = document.getElementById("btnOpenFlashcardModal");
  const modal = document.getElementById("flashcardModal");
  const btnClose = document.getElementById("btnCloseFcModal");
  const btnCancel = document.getElementById("btnCancelFc");
  const form = document.getElementById("flashcardForm");

  const toggleFlip = () => cardInner.classList.toggle("flipped");

  if (cardInner) cardInner.addEventListener("click", toggleFlip);
  if (btnFlip) btnFlip.addEventListener("click", toggleFlip);

  window.addEventListener("keydown", (e) => {
    if (e.code === "Space" && document.getElementById("tab-flashcards")?.classList.contains("active") && !document.qu
erySelector(".modal-backdrop.active")) {
      e.preventDefault();
      toggleFlip();
    }
  });

  if (btnPrev) {
    btnPrev.addEventListener("click", () => {
      if (dbMgr.cache.flashcards.length === 0) return;
      cardInner.classList.remove("flipped");
      currentCardIndex = (currentCardIndex - 1 + dbMgr.cache.flashcards.length) % dbMgr.cache.flashcards.length;
      renderActiveCard();
    });
  }

  if (btnNext) {
    btnNext.addEventListener("click", () => {
      if (dbMgr.cache.flashcards.length === 0) return;
      cardInner.classList.remove("flipped");
      currentCardIndex = (currentCardIndex + 1) % dbMgr.cache.flashcards.length;
      renderActiveCard();
    });
  }

  if (btnMaster) {
```

```
  btnMaster.addEventListener("click", () => {
    if (dbMgr.cache.flashcards.length > 0) {
      const c = dbMgr.cache.flashcards[currentCardIndex];
      dbMgr.toggleFlashcardMastery(c.id);
      renderActiveCard();
    }
  });
}

if (btnOpenModal) {
  btnOpenModal.addEventListener("click", () => {
    updateSubjectSelectors();
    modal.classList.add("active");
  });
}
const closeFcModal = () => modal.classList.remove("active");
if (btnClose) btnClose.addEventListener("click", closeFcModal);
if (btnCancel) btnCancel.addEventListener("click", closeFcModal);

if (form) {
  form.addEventListener("submit", (e) => {
    e.preventDefault();
    const topicSubject = document.getElementById("fcSubjectInput").value.trim();
    const newCard = {
      id: "FC-" + (Date.now() % 100000),
      subject: topicSubject,
      question: document.getElementById("fcQuestionInput").value.trim(),
      answer: document.getElementById("fcAnswerInput").value.trim(),
      mastered: false
    };
    dbMgr.addFlashcard(newCard);

    if (topicSubject) {
      dbMgr.addSubject(topicSubject);
      updateSubjectSelectors();
    }

    currentCardIndex = dbMgr.cache.flashcards.length - 1;
    cardInner.classList.remove("flipped");
    renderActiveCard();
    form.reset();
    closeFcModal();
  });
}
```

```
  renderActiveCard();
}

function renderActiveCard() {
  const fcSubject = document.getElementById("fcSubject");
  const fcCounter = document.getElementById("fcCounter");
  const fcQuestion = document.getElementById("fcQuestion");
  const fcAnswer = document.getElementById("fcAnswer");
  const fcMasteryTag = document.getElementById("fcMasteryTag");
  const fcMasterBtnText = document.getElementById("fcMasterBtnText");

  if (dbMgr.cache.flashcards.length === 0) {
    if (fcSubject) fcSubject.textContent = "Active Recall";
    if (fcCounter) fcCounter.textContent = "0 Cards";
    if (fcQuestion) fcQuestion.textContent = "No revision flashcards added yet. Click '+ New Card' to add concept pro
mpts for active recall.";
    if (fcAnswer) fcAnswer.textContent = "Answers will appear here once cards are created.";
    if (fcMasteryTag) fcMasteryTag.textContent = "Empty";
    if (fcMasterBtnText) fcMasterBtnText.textContent = "Mark Mastered";
    return;
  }

  const card = dbMgr.cache.flashcards[currentCardIndex];
  if (!card) return;

  if (fcSubject) fcSubject.textContent = card.subject;
  if (fcCounter) fcCounter.textContent = `Card ${currentCardIndex + 1} of ${dbMgr.cache.flashcards.length}`;
  if (fcQuestion) fcQuestion.textContent = card.question;
  if (fcAnswer) fcAnswer.textContent = card.answer;

  if (card.mastered) {
    if (fcMasteryTag) {
      fcMasteryTag.textContent = "Mastered";
      fcMasteryTag.style.color = "var(--accent)";
    }
    if (fcMasterBtnText) fcMasterBtnText.textContent = "Mastered";
  } else {
    if (fcMasteryTag) {
      fcMasteryTag.textContent = "Unmastered";
      fcMasteryTag.style.color = "var(--text-muted)";
    }
    if (fcMasterBtnText) fcMasterBtnText.textContent = "Mark Mastered";
  }
}

// ========================================================================
```

```
// Module 5: Analytics & Data Export
// ================================================================================
function initAnalytics() {
  const btnExportTxt = document.getElementById("btnExportTxt");
  const btnExportJson = document.getElementById("btnExportJson");

  if (btnExportTxt) btnExportTxt.addEventListener("click", downloadTextReport);
  if (btnExportJson) btnExportJson.addEventListener("click", downloadJsonBackup);
}

function renderAnalytics() {
  const statCourseCount = document.getElementById("statCourseCount");
  const statTaskCount = document.getElementById("statTaskCount");
  const statMasteredCount = document.getElementById("statMasteredCount");
  const statStudyMins = document.getElementById("statStudyMins");
  const reportPreviewBox = document.getElementById("reportPreviewBox");

  if (statCourseCount) statCourseCount.textContent = dbMgr.cache.courses.length;
  if (statTaskCount) statTaskCount.textContent = dbMgr.cache.tasks.length;

  const mastered = dbMgr.cache.flashcards.filter(f => f.mastered).length;
  if (statMasteredCount) statMasteredCount.textContent = `${mastered} / ${dbMgr.cache.flashcards.length}`;

  if (statStudyMins) statStudyMins.textContent = `${dbMgr.cache.pomodoroStats.totalMinutes}m`;

  const report = generateReportText();
  if (reportPreviewBox) reportPreviewBox.textContent = report;
}

function generateReportText() {
  const { sgpa, totalCredits } = calculateSgpa();
  const dateStr = new Date().toISOString().split('T')[0];

  return `================================================================================
                  STUDENT WORKSPACE • ACADEMIC REPORT
================================================================================
Generated Date:   ${dateStr}
Workspace:        Student Workspace
Current SGPA:     ${sgpa.toFixed(2)} / 10.00
Enrolled Credits: ${totalCredits.toFixed(1)} Credits
Study Time:       ${dbMgr.cache.pomodoroStats.totalMinutes} minutes (${dbMgr.cache.pomodoroStats.completedSessions} f
ocus blocks)
Database:         SQLite WebAssembly Engine (Local Client-Side Storage)
================================================================================

1. REGISTERED SEMESTER COURSES:
```

```
--------------------------------------------------------------------------------
Code          Course Title                          Credits   Grade   Points
--------------------------------------------------------------------------------
${dbMgr.cache.courses.length === 0 ? "No courses recorded yet." : dbMgr.cache.courses.map(c => {
  const pt = (GRADE_POINTS[c.grade] ?? 0.0).toFixed(1);
  return `${c.code.padEnd(13)} ${c.name.padEnd(41)} ${c.credits.toFixed(1).padEnd(9)} ${c.grade.padEnd(7)} ${pt}`;
}).join('\n')}
--------------------------------------------------------------------------------

2. ASSIGNMENTS & DELIVERABLES:
--------------------------------------------------------------------------------
Title                                 Course                 Due Date     Priority
--------------------------------------------------------------------------------
${dbMgr.cache.tasks.length === 0 ? "No assignments recorded yet." : dbMgr.cache.tasks.map(t => {
  const status = t.completed ? "[DONE]" : "[DUE] ";
  return `${status} ${t.title.padEnd(35)} ${t.course.padEnd(23)} ${t.dueDate.padEnd(12)} ${t.priority}`;
}).join('\n')}
--------------------------------------------------------------------------------

3. FLASHCARDS RETENTION:
Total Cards:      ${dbMgr.cache.flashcards.length}
Mastered Topics:  ${dbMgr.cache.flashcards.filter(f => f.mastered).length}
================================================================================`;
}

function downloadTextReport() {
  const report = generateReportText();
  downloadBlob(report, "Student_Workspace_Academic_Report.txt", "text/plain");
}

function downloadJsonBackup() {
  const jsonStr = JSON.stringify(dbMgr.cache, null, 2);
  downloadBlob(jsonStr, "student_workspace_backup.json", "application/json");
}

function downloadBlob(content, filename, contentType) {
  const blob = new Blob([content], { type: contentType });
  const url = URL.createObjectURL(blob);
  const a = document.createElement("a");
  a.href = url;
  a.download = filename;
  document.body.appendChild(a);
  a.click();
  document.body.removeChild(a);
  URL.revokeObjectURL(url);
}

function escapeHtml(str) {
  if (!str) return "";
  return str.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;").replace(/"/g, "&quot;");
}
```

## 3. Styling: style.css (Calm Scholarly Design System & Pastel Accents)

```
/* ========================================================================
   Student Productivity Suite • Calm Scholarly Aesthetic
   Crafted following the Unslop specification (Damir Sokolovsky, CC BY 4.0)
   ======================================================================== */

:root {
  /* Warm Paper & Ink Neutral Ramp */
  --bg-canvas: #F7F5F0;
  --bg-surface: #FFFFFF;
  --bg-raised: #EFEBE3;
  --bg-subtle: #F2EFE9;

  --border-hairline: rgba(31, 27, 22, 0.08);
  --border-subtle: rgba(31, 27, 22, 0.12);
  --border-focus: #2D6A4F;

  --text-primary: #1F1B16;
  --text-secondary: #5E554A;
  --text-muted: #7B7266;

  /* Single Accent: Academic Forest Green */
  --accent: #2D6A4F;
  --accent-hover: #1E4620;
  --accent-tint: #EBF5EE;

  /* Semantic Tags (Toned down) */
  --tag-high-bg: #FDF2E9;
  --tag-high-text: #9A3412;
  --tag-med-bg: #FEF9C3;
  --tag-med-text: #854D0E;
  --tag-low-bg: #F1F5F9;
  --tag-low-text: #475569;
  --tag-done-bg: #EBF7EE;
  --tag-done-text: #166534;

  /* Typography */
  --font-sans: 'Plus Jakarta Sans', -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif;
  --font-mono: 'JetBrains Mono', ui-monospace, SFMono-Regular, monospace;

  /* Nested Radii */
  --radius-card: 12px;
  --radius-control: 6px;
  --radius-tag: 4px;

  /* Depth */
```

```
  --shadow-card: 0 1px 2px rgba(31, 27, 22, 0.04), 0 1px 1px rgba(31, 27, 22, 0.02);
  --shadow-float: 0 8px 20px rgba(31, 27, 22, 0.06);

  /* Easing */
  --ease: cubic-bezier(0.22, 1, 0.36, 1);
  --duration: 140ms;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: var(--font-sans);
  background-color: var(--bg-canvas);
  color: var(--text-primary);
  line-height: 1.5;
  -webkit-font-smoothing: antialiased;
}

.app-layout {
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

/* ============================================================================
   Header Bar
   ============================================================================ */
.app-header {
  background-color: var(--bg-surface);
  border-bottom: 1px solid var(--border-hairline);
  height: 56px;
  padding: 0 24px;
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.header-left {
  display: flex;
  align-items: center;
}
```

```
.brand-title {
  font-size: 1.15rem;
  font-weight: 700;
  color: var(--text-primary);
  letter-spacing: -0.02em;
}

.header-right {
  display: flex;
  align-items: center;
  gap: 14px;
}

.status-metric {
  display: flex;
  align-items: baseline;
  gap: 6px;
}

.metric-label {
  font-size: 0.72rem;
  font-weight: 500;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.03em;
}

.metric-value {
  font-family: var(--font-mono);
  font-size: 0.85rem;
  font-weight: 600;
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.status-divider {
  width: 1px;
  height: 14px;
  background-color: var(--border-subtle);
}

/* ================================================================================
   Navigation Bar
   ================================================================================ */
```

```
.app-nav {
  background-color: var(--bg-surface);
  border-bottom: 1px solid var(--border-hairline);
  padding: 0 24px;
}

.nav-container {
  max-width: 1200px;
  margin: 0;
}

.nav-tabs {
  display: flex;
  gap: 6px;
  justify-content: flex-start;
  align-items: center;
  overflow-x: auto;
  scrollbar-width: none;
  -webkit-overflow-scrolling: touch;
}

.nav-tabs::-webkit-scrollbar {
  display: none;
}

.nav-tab {
  background: none;
  border: none;
  height: 42px;
  padding: 0 16px;
  font-family: inherit;
  font-size: 0.875rem;
  font-weight: 500;
  color: var(--text-secondary);
  cursor: pointer;
  border-bottom: 2px solid transparent;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  transition: color var(--duration) var(--ease), border-color var(--duration) var(--ease);
  white-space: nowrap;
  flex-shrink: 0;
}

.nav-tab:hover {
```

```
  color: var(--text-primary);
}

.nav-tab.active {
  color: var(--accent);
  border-bottom-color: var(--accent);
  font-weight: 600;
}

.tab-icon {
  color: currentColor;
}

/* ================================================================================
   Main Layout & Structure
   ================================================================================ */
.app-main {
  flex: 1;
  padding: 32px 24px;
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
}

.tab-pane {
  display: none;
}

.tab-pane.active {
  display: block;
}

.section-header {
  display: flex;
  align-items: flex-end;
  justify-content: space-between;
  margin-bottom: 24px;
}

.section-title {
  font-size: 1.35rem;
  font-weight: 600;
  letter-spacing: -0.015em;
  color: var(--text-primary);
}
```

```
.section-desc {
  font-size: 0.85rem;
  color: var(--text-muted);
  margin-top: 2px;
}

.section-actions {
  display: flex;
  gap: 8px;
}

/* ================================================================================
   Card Component
   ================================================================================ */
.card {
  background-color: var(--bg-surface);
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-card);
  padding: 20px;
  box-shadow: var(--shadow-card);
}

.card-title {
  font-size: 0.95rem;
  font-weight: 600;
  color: var(--text-primary);
  margin-bottom: 12px;
}

.card-caption {
  font-size: 0.75rem;
  font-weight: 500;
  color: var(--text-muted);
}

/* ================================================================================
   Buttons
   ================================================================================ */
.btn {
  font-family: inherit;
  font-size: 0.825rem;
  font-weight: 500;
  height: 32px;
  padding: 0 12px;
```

```
  border-radius: var(--radius-control);
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  transition: background-color var(--duration) var(--ease), border-color var(--duration) var(--ease), color var(--dur
ation) var(--ease);
  border: 1px solid transparent;
}

.btn-primary {
  background-color: var(--accent);
  color: white;
  border-color: var(--accent);
}
.btn-primary:hover {
  background-color: var(--accent-hover);
  border-color: var(--accent-hover);
}

.btn-secondary {
  background-color: var(--bg-surface);
  border-color: var(--border-subtle);
  color: var(--text-primary);
}
.btn-secondary:hover {
  background-color: var(--bg-raised);
}

.btn-ghost {
  background: none;
  color: var(--text-muted);
}
.btn-ghost:hover {
  background-color: var(--bg-subtle);
  color: var(--text-primary);
}

.btn-md {
  height: 36px;
  padding: 0 16px;
  font-size: 0.85rem;
}

.btn-sm {
```

```
  height: 26px;
  padding: 0 8px;
  font-size: 0.75rem;
}

.btn:disabled {
  opacity: 0.45;
  cursor: not-allowed;
}

/* ================================================================================
   Tags & Badges
   ================================================================================ */
.status-tag {
  font-size: 0.75rem;
  font-weight: 500;
  padding: 2px 6px;
  border-radius: var(--radius-tag);
  display: inline-block;
}

.tag-green {
  background-color: var(--tag-done-bg);
  color: var(--tag-done-text);
}

.tag-neutral {
  background-color: var(--bg-raised);
  color: var(--text-secondary);
}

.tag-category {
  font-size: 0.75rem;
  font-weight: 500;
  color: var(--accent);
  background-color: var(--accent-tint);
  padding: 2px 6px;
  border-radius: var(--radius-tag);
}

/* ================================================================================
   TAB 1: ASSIGNMENTS
   ================================================================================ */
.filter-panel {
  background-color: var(--bg-surface);
```

```
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-card);
  padding: 10px 14px;
  margin-bottom: 20px;
  display: flex;
  align-items: center;
  gap: 16px;
  flex-wrap: wrap;
}

.search-field {
  flex: 1;
  min-width: 240px;
  position: relative;
  display: flex;
  align-items: center;
}

.field-icon {
  position: absolute;
  left: 10px;
  color: var(--text-muted);
}

.search-field input {
  width: 100%;
  height: 32px;
  padding: 0 10px 0 32px;
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-control);
  font-family: inherit;
  font-size: 0.85rem;
  background-color: var(--bg-canvas);
  outline: none;
  transition: border-color var(--duration) var(--ease);
}
.search-field input:focus {
  border-color: var(--border-focus);
  background-color: white;
}

.filter-controls {
  display: flex;
  align-items: center;
  gap: 12px;
```

```
}

.select-wrapper {
  display: flex;
  align-items: center;
  gap: 6px;
}

.select-label {
  font-size: 0.75rem;
  color: var(--text-muted);
}

.form-select-sm {
  height: 30px;
  padding: 0 8px;
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-control);
  font-family: inherit;
  font-size: 0.8rem;
  background-color: white;
  outline: none;
}

.task-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(340px, 1fr));
  gap: 16px;
}

.task-card {
  background-color: var(--bg-surface);
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-card);
  padding: 16px;
  display: flex;
  flex-direction: column;
  gap: 12px;
  box-shadow: var(--shadow-card);
  transition: border-color var(--duration) var(--ease);
}
.task-card:hover {
  border-color: var(--border-subtle);
}
```

```
.task-card.completed {
  opacity: 0.55;
  background-color: var(--bg-canvas);
}

.task-card-header {
  display: flex;
  align-items: flex-start;
  justify-content: space-between;
  gap: 12px;
}

.task-title {
  font-size: 0.925rem;
  font-weight: 600;
  color: var(--text-primary);
  line-height: 1.35;
}

.task-course {
  font-size: 0.785rem;
  color: var(--text-muted);
  margin-top: 2px;
}

.priority-tag {
  font-size: 0.7rem;
  font-weight: 500;
  padding: 2px 6px;
  border-radius: var(--radius-tag);
}

.priority-high {
  background-color: var(--tag-high-bg);
  color: var(--tag-high-text);
}
.priority-medium {
  background-color: var(--tag-med-bg);
  color: var(--tag-med-text);
}
.priority-low {
  background-color: var(--tag-low-bg);
  color: var(--tag-low-text);
}
```

```
.task-meta-row {
  display: flex;
  align-items: center;
  justify-content: space-between;
  font-size: 0.775rem;
  color: var(--text-muted);
  border-top: 1px solid var(--border-hairline);
  padding-top: 10px;
}

.task-footer-actions {
  display: flex;
  align-items: center;
  justify-content: space-between;
}

/* ================================================================================
   TAB 2: SGPA CALCULATOR
   ================================================================================ */
.gpa-overview-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 16px;
  margin-bottom: 24px;
}

.gpa-metric-card {
  display: flex;
  flex-direction: column;
  justify-content: space-between;
}

.metric-top {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 8px;
}

.gpa-display-row {
  display: flex;
  align-items: baseline;
  gap: 6px;
  margin: 12px 0;
}
```

```
.gpa-large-number {
  font-family: var(--font-mono);
  font-size: 2.8rem;
  font-weight: 500;
  line-height: 1;
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.gpa-max-scale {
  font-size: 0.85rem;
  color: var(--text-muted);
}

.gpa-breakdown-row {
  display: flex;
  gap: 24px;
  border-top: 1px solid var(--border-hairline);
  padding-top: 12px;
}

.breakdown-label {
  display: block;
  font-size: 0.75rem;
  color: var(--text-muted);
}

.breakdown-val {
  font-family: var(--font-mono);
  font-size: 0.95rem;
  font-weight: 500;
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.gpa-simulator-card {
  display: flex;
  flex-direction: column;
  justify-content: space-between;
}

.simulator-intro {
  font-size: 0.8rem;
  color: var(--text-secondary);
```

```
  margin-bottom: 12px;
}

.sim-controls {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-bottom: 12px;
}

.input-inline {
  display: flex;
  align-items: center;
  gap: 8px;
}

.input-inline label {
  font-size: 0.775rem;
  color: var(--text-muted);
}

.input-inline input {
  width: 76px;
  height: 32px;
  padding: 0 8px;
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-control);
  font-family: var(--font-mono);
  font-size: 0.85rem;
  text-align: center;
}

.sim-result-box {
  background-color: var(--bg-canvas);
  border-left: 2px solid var(--accent);
  padding: 10px 12px;
  border-radius: var(--radius-tag);
  font-size: 0.8rem;
  color: var(--text-secondary);
}

.table-card {
  padding: 0;
  overflow: hidden;
}
```

```
.data-table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.825rem;
  text-align: left;
}

.data-table th {
  background-color: var(--bg-raised);
  height: 36px;
  padding: 0 16px;
  font-weight: 500;
  color: var(--text-secondary);
  border-bottom: 1px solid var(--border-hairline);
}

.data-table td {
  height: 40px;
  padding: 0 16px;
  border-bottom: 1px solid var(--border-hairline);
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.data-table tr:last-child td {
  border-bottom: none;
}

.code-pill {
  font-family: var(--font-mono);
  font-size: 0.75rem;
  background-color: var(--bg-subtle);
  padding: 2px 6px;
  border-radius: var(--radius-tag);
  border: 1px solid var(--border-hairline);
}

/* ================================================================================
   TAB 3: FOCUS TIMER
   ================================================================================ */
.pomodoro-grid {
  display: grid;
  grid-template-columns: 1.15fr 0.85fr;
  gap: 20px;
```

```
}

.timer-card {
  display: flex;
  flex-direction: column;
  align-items: center;
  padding: 32px 24px;
}

.mode-selector {
  display: flex;
  gap: 4px;
  background-color: var(--bg-raised);
  padding: 3px;
  border-radius: var(--radius-control);
}

.mode-btn {
  background: none;
  border: none;
  height: 28px;
  padding: 0 12px;
  border-radius: var(--radius-tag);
  font-family: inherit;
  font-size: 0.775rem;
  font-weight: 500;
  color: var(--text-muted);
  cursor: pointer;
  transition: all var(--duration) var(--ease);
}

.mode-btn.active {
  background-color: white;
  color: var(--text-primary);
  box-shadow: 0 1px 2px rgba(31, 27, 22, 0.05);
}

.dial-container {
  position: relative;
  width: 240px;
  height: 240px;
  margin: 28px 0;
  display: flex;
  align-items: center;
  justify-content: center;
```

```
}

.dial-svg {
  transform: rotate(-90deg);
}

.dial-track {
  fill: none;
  stroke: var(--bg-raised);
  stroke-width: 6;
}

.dial-indicator {
  fill: none;
  stroke: var(--accent);
  stroke-width: 6;
  stroke-linecap: round;
  stroke-dasharray: 653.45;
  stroke-dashoffset: 0;
  transition: stroke-dashoffset 0.5s ease;
}

.dial-center {
  position: absolute;
  display: flex;
  flex-direction: column;
  align-items: center;
}

.timer-readout {
  font-family: var(--font-mono);
  font-size: 3rem;
  font-weight: 500;
  letter-spacing: -0.04em;
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.timer-status-text {
  font-size: 0.8rem;
  color: var(--text-muted);
}

.timer-action-row {
  display: flex;
```

```
  gap: 8px;
  margin-bottom: 24px;
}

.timer-meta-strip {
  width: 100%;
  display: flex;
  align-items: center;
  justify-content: space-around;
  border-top: 1px solid var(--border-hairline);
  padding-top: 18px;
}

.meta-stat {
  text-align: center;
}

.meta-num {
  font-family: var(--font-mono);
  font-size: 1.15rem;
  font-weight: 500;
  color: var(--text-primary);
  display: block;
}

.meta-lbl {
  font-size: 0.725rem;
  color: var(--text-muted);
}

.meta-stat-divider {
  width: 1px;
  height: 24px;
  background-color: var(--border-hairline);
}

.timer-side-column {
  display: flex;
  flex-direction: column;
  gap: 16px;
}

.guideline-list {
  list-style: none;
  display: flex;
```

```
  flex-direction: column;
  gap: 14px;
}

.guideline-list li {
  display: flex;
  align-items: flex-start;
  gap: 12px;
  font-size: 0.8rem;
  color: var(--text-secondary);
}

.guideline-list li p {
  color: var(--text-muted);
  margin-top: 2px;
}

.step-num {
  font-family: var(--font-mono);
  font-size: 0.725rem;
  width: 20px;
  height: 20px;
  background-color: var(--bg-raised);
  border-radius: var(--radius-tag);
  display: flex;
  align-items: center;
  justify-content: center;
  color: var(--text-secondary);
  flex-shrink: 0;
}

/* ================================================================================
   TAB 4: FLASHCARDS
   ================================================================================ */
.flashcard-deck-layout {
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 20px;
}

.flashcard-viewport {
  width: 100%;
  max-width: 640px;
  height: 340px;
```

```
  perspective: 1000px;
  cursor: pointer;
}

.flashcard-box {
  position: relative;
  width: 100%;
  height: 100%;
  transition: transform 400ms var(--ease);
  transform-style: preserve-3d;
}

.flashcard-box.flipped {
  transform: rotateY(180deg);
}

.card-side {
  position: absolute;
  width: 100%;
  height: 100%;
  backface-visibility: hidden;
  background-color: var(--bg-surface);
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-card);
  padding: 24px;
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  box-shadow: var(--shadow-card);
}

.side-back {
  transform: rotateY(180deg);
  background-color: #FAF9F6;
}

.side-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
}

.side-counter, .mastery-indicator {
  font-size: 0.75rem;
  color: var(--text-muted);
```

```
}

.content-eyebrow {
  display: block;
  font-size: 0.725rem;
  font-weight: 600;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.04em;
  margin-bottom: 8px;
}

.content-text {
  font-size: 1.15rem;
  font-weight: 500;
  line-height: 1.5;
  color: var(--text-primary);
}

.side-footer {
  font-size: 0.75rem;
  color: var(--text-muted);
  text-align: center;
  border-top: 1px dashed var(--border-hairline);
  padding-top: 12px;
}

.keycap {
  font-family: var(--font-mono);
  font-size: 0.7rem;
  background-color: var(--bg-raised);
  padding: 1px 5px;
  border-radius: var(--radius-tag);
  border: 1px solid var(--border-subtle);
  color: var(--text-secondary);
}

.flashcard-nav-row {
  display: flex;
  gap: 8px;
}

/* ================================================================================
   TAB 5: ANALYTICS & EXPORT
   ================================================================================ */
```

```
.analytics-stat-row {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 16px;
  margin-bottom: 24px;
}

.stat-card {
  padding: 16px;
}

.stat-caption {
  font-size: 0.75rem;
  color: var(--text-muted);
  display: block;
  margin-bottom: 6px;
}

.stat-metric-large {
  font-family: var(--font-mono);
  font-size: 1.6rem;
  font-weight: 500;
  color: var(--text-primary);
  font-feature-settings: "tnum" 1;
}

.report-box {
  padding: 20px;
}

.report-box-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 12px;
}

.ascii-preview {
  background-color: #1A1A18;
  color: #EDECE8;
  font-family: var(--font-mono);
  font-size: 0.775rem;
  padding: 16px;
  border-radius: var(--radius-control);
  overflow-x: auto;
```

```
  line-height: 1.5;
}

/* ================================================================================
   Modals
   ================================================================================ */
.modal-backdrop {
  position: fixed;
  inset: 0;
  background-color: rgba(31, 27, 22, 0.4);
  backdrop-filter: blur(2px);
  display: none;
  align-items: center;
  justify-content: center;
  padding: 20px;
  z-index: 100;
}

.modal-backdrop.active {
  display: flex;
}

.modal-dialog {
  background-color: var(--bg-surface);
  border-radius: var(--radius-card);
  padding: 24px;
  width: 100%;
  max-width: 460px;
  box-shadow: var(--shadow-float);
  border: 1px solid var(--border-hairline);
}

.modal-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 18px;
}

.modal-title {
  font-size: 1.05rem;
  font-weight: 600;
}

.btn-close {
```

```
  background: none;
  border: none;
  font-size: 1rem;
  color: var(--text-muted);
  cursor: pointer;
  padding: 4px;
}

.form-group {
  margin-bottom: 14px;
}

.form-group label {
  display: block;
  font-size: 0.775rem;
  font-weight: 500;
  color: var(--text-secondary);
  margin-bottom: 4px;
}

.form-group input, .form-group select, .form-group textarea, .form-select {
  width: 100%;
  height: 34px;
  padding: 0 10px;
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-control);
  font-family: inherit;
  font-size: 0.825rem;
  background-color: white;
  outline: none;
}

.form-group textarea {
  height: auto;
  padding: 8px 10px;
}

.form-group input:focus, .form-group select:focus, .form-group textarea:focus, .form-select:focus {
  border-color: var(--border-focus);
}

.form-row-2 {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 12px;
```

```
}

.modal-actions {
  display: flex;
  justify-content: flex-end;
  gap: 8px;
  margin-top: 20px;
}

/* ================================================================================
   Timer Duration Controls & Subject Management
   ================================================================================ */
.duration-editor-strip {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 12px;
  margin-bottom: 20px;
  padding: 8px 14px;
  background-color: var(--bg-subtle);
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-control);
  flex-wrap: wrap;
}

.duration-editor-label {
  font-size: 0.8rem;
  font-weight: 500;
  color: var(--text-secondary);
}

.duration-controls-row {
  display: flex;
  align-items: center;
  gap: 8px;
}

.duration-input {
  width: 68px;
  height: 32px;
  padding: 0 8px;
  font-family: var(--font-mono);
  font-size: 0.9rem;
  font-weight: 600;
  text-align: center;
```

```
  border: 1px solid var(--border-hairline);
  border-radius: var(--radius-sm);
  background: #ffffff;
  color: var(--text-primary);
}

.duration-input:focus {
  outline: none;
  border-color: var(--accent);
}

.duration-unit {
  font-size: 0.8rem;
  color: var(--text-muted);
}

.add-subject-panel {
  margin-top: 14px;
  padding-top: 14px;
  border-top: 1px solid var(--border-hairline);
}

.add-subject-panel label {
  display: block;
  font-size: 0.775rem;
  font-weight: 500;
  color: var(--text-muted);
  margin-bottom: 6px;
}

.add-subject-input-row {
  display: flex;
  gap: 8px;
}

.add-subject-input-row input {
  flex: 1;
  font-size: 0.85rem;
}

.field-hint {
  font-size: 0.725rem;
  color: var(--text-muted);
  margin-top: 6px;
  line-height: 1.35;
```

```
}

.table-scroll-container {
  width: 100%;
  overflow-x: auto;
  -webkit-overflow-scrolling: touch;
}

/* Empty States */
.empty-state-card {
  grid-column: 1 / -1;
  padding: 3rem 1.5rem;
  text-align: center;
  background: var(--surface);
  border: 1px dashed var(--border);
  border-radius: var(--radius-md);
  color: var(--text-muted);
  font-size: 0.9rem;
}

.empty-state-cell {
  text-align: center !important;
  padding: 2.5rem 1rem !important;
  color: var(--text-muted) !important;
  font-size: 0.88rem;
}

.empty-state-action {
  margin-top: 10px;
}

/* ================================================================================
   Responsive Breakpoints: Desktop, Tablet & Mobile Screens
   ================================================================================ */

/* Tablet & iPad screens (max-width: 1024px) */
@media (max-width: 1024px) {
  .app-main {
    padding: 24px 18px;
    max-width: 100%;
  }
  .gpa-overview-grid {
    grid-template-columns: 1fr;
    gap: 16px;
  }
```

```
  .pomodoro-grid {
    grid-template-columns: 1fr;
    gap: 16px;
  }
  .analytics-stat-row {
    grid-template-columns: repeat(2, 1fr);
    gap: 12px;
  }
}

/* Mobile & Small Tablet screens (max-width: 640px) */
@media (max-width: 640px) {
  .app-header {
    height: auto;
    min-height: 52px;
    padding: 10px 14px;
    flex-direction: row;
    flex-wrap: wrap;
    justify-content: space-between;
    gap: 8px;
  }
  .brand-title {
    font-size: 1.05rem;
  }
  .header-right {
    gap: 8px;
  }
  .status-metric {
    gap: 4px;
  }
  .metric-label {
    font-size: 0.68rem;
  }
  .metric-value {
    font-size: 0.78rem;
  }
  .status-divider {
    height: 11px;
  }
  .app-nav {
    padding: 0 12px;
  }
  .nav-tabs {
    gap: 4px;
    padding: 2px 0;
```

```
}
.nav-tab {
  height: 38px;
  padding: 0 12px;
  font-size: 0.8rem;
}
.app-main {
  padding: 16px 10px;
}
.section-header {
  flex-direction: column;
  align-items: flex-start;
  gap: 10px;
}
.section-title {
  font-size: 1.35rem;
}
.section-actions {
  width: 100%;
  display: flex;
  gap: 8px;
}
.section-actions .btn {
  flex: 1;
  justify-content: center;
}
.filter-panel {
  flex-direction: column;
  gap: 8px;
}
.search-field {
  width: 100%;
}
.filter-controls {
  width: 100%;
  display: flex;
  gap: 8px;
}
.filter-controls .select-wrapper {
  flex: 1;
}
.filter-controls select {
  width: 100%;
}
.task-grid {
```

```
    grid-template-columns: 1fr;
    gap: 12px;
  }
  .mode-selector {
    display: flex;
    flex-wrap: wrap;
    gap: 4px;
    justify-content: center;
  }
  .mode-btn {
    flex: 1 1 auto;
    padding: 6px 10px;
    font-size: 0.75rem;
    text-align: center;
  }
  .duration-editor-strip {
    flex-direction: column;
    gap: 8px;
    padding: 10px;
  }
  .timer-action-row {
    display: flex;
    gap: 8px;
    width: 100%;
  }
  .timer-action-row .btn {
    flex: 1;
    justify-content: center;
  }
  .timer-meta-strip {
    gap: 8px;
    flex-wrap: wrap;
  }
  .meta-stat {
    min-width: 60px;
  }
  .meta-num {
    font-size: 1rem;
  }
  .meta-lbl {
    font-size: 0.65rem;
  }
  .flashcard-nav-row {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 8px;
    width: 100%;
  }
  .flashcard-nav-row .btn {
    width: 100%;
    justify-content: center;
  }
  .analytics-stat-row {
    grid-template-columns: 1fr;
    gap: 8px;
  }
  .modal-dialog {
    width: calc(100% - 20px);
    margin: 10px auto;
    padding: 16px 14px;
    border-radius: var(--radius-card);
  }
  .form-row-2 {
    grid-template-columns: 1fr;
    gap: 8px;
  }
}
```

## 4. Database Layer: lib/db.js (WebAssembly sql.js SQLite Engine)

```
import initSqlJs from "sql.js";
import fs from "fs";
import path from "path";

let SQL = null;
let dbInstance = null;

// File persistence paths
const isVercel = !!process.env.VERCEL;
const dbDir = isVercel ? "/tmp" : path.join(process.cwd(), "data");

if (!fs.existsSync(dbDir)) {
  try {
    fs.mkdirSync(dbDir, { recursive: true });
  } catch (e) {}
}

const dbPath = path.join(dbDir, "student_suite.db");

function persistDb() {
  if (!dbInstance) return;
  try {
    const data = dbInstance.export();
    const buffer = Buffer.from(data);
    fs.writeFileSync(dbPath, buffer);
  } catch (err) {
    console.error("Failed to persist SQLite database to disk:", err);
  }
}

export async function getDb() {
  if (!SQL) {
    SQL = await initSqlJs();
  }

  if (!dbInstance) {
    // Check if seeded database file already exists
    if (fs.existsSync(dbPath)) {
      try {
        const buffer = fs.readFileSync(dbPath);
        dbInstance = new SQL.Database(buffer);
      } catch (e) {
        dbInstance = new SQL.Database();
      }
    } else {
```

```
      dbInstance = new SQL.Database();
    }

    initSchema(dbInstance);
  }

  return {
    query: (sql, params = []) => {
      const stmt = dbInstance.prepare(sql);
      stmt.bind(params);
      const rows = [];
      while (stmt.step()) {
        rows.push(stmt.getAsObject());
      }
      stmt.free();
      return rows;
    },
    run: (sql, params = []) => {
      dbInstance.run(sql, params);
      persistDb();
    }
  };
}

function initSchema(db) {
  // 1. Courses Table
  db.run(`
    CREATE TABLE IF NOT EXISTS courses (
      id TEXT PRIMARY KEY,
      code TEXT NOT NULL,
      name TEXT NOT NULL,
      credits REAL NOT NULL,
      grade TEXT NOT NULL,
      grade_point REAL NOT NULL,
      created_at INTEGER NOT NULL
    )
  `);

  // 2. Tasks Table
  db.run(`
    CREATE TABLE IF NOT EXISTS tasks (
      id TEXT PRIMARY KEY,
      title TEXT NOT NULL,
      course TEXT NOT NULL,
      due_date TEXT NOT NULL,
      priority TEXT NOT NULL,
      hours INTEGER NOT NULL,
      completed INTEGER NOT NULL DEFAULT 0,
      created_at INTEGER NOT NULL
    )
  `);

  // 3. Flashcards Table
  db.run(`
    CREATE TABLE IF NOT EXISTS flashcards (
      id TEXT PRIMARY KEY,
      subject TEXT NOT NULL,
      question TEXT NOT NULL,
      answer TEXT NOT NULL,
      mastered INTEGER NOT NULL DEFAULT 0,
      created_at INTEGER NOT NULL
    )
  `);

  // 4. Pomodoro Sessions Table
  db.run(`
    CREATE TABLE IF NOT EXISTS pomodoro_sessions (
      id TEXT PRIMARY KEY,
      subject TEXT NOT NULL,
      duration_minutes INTEGER NOT NULL,
      completed_at INTEGER NOT NULL
    )
  `);

  // Empty schema initialized without pre-seeding
  persistDb();
}
```

## 5. React Frontend: pages/index.js (Next.js Page & SQLite Integration)

```
import React, { useState, useEffect, useMemo, useRef } from "react";
import Head from "next/head";
import { getDb } from "../lib/db";

const GRADE_POINTS = {
  "A+": 10.0,
  "A": 9.0,
  "B+": 8.0,
  "B": 7.0,
  "C+": 6.0,
  "C": 5.0,
  "D": 4.0,
  "F": 0.0
};

export default function Home({ initialData }) {
  // Local state initialized from SQLite Database
  const [activeTab, setActiveTab] = useState("assignments");
  const [courses, setCourses] = useState(initialData?.courses || []);
  const [tasks, setTasks] = useState(initialData?.tasks || []);
  const [flashcards, setFlashcards] = useState(initialData?.flashcards || []);
  const [pomodoroStats, setPomodoroStats] = useState(initialData?.pomodoro || { completedSessions: 4, totalMinutes: 1
00, streakDays: 4 });

  // Filtering & search
  const [taskSearch, setTaskSearch] = useState("");
  const [priorityFilter, setPriorityFilter] = useState("ALL");
  const [statusFilter, setStatusFilter] = useState("ALL");

  // GPA Simulator
  const [simTarget, setSimTarget] = useState("9.50");
  const [simMessage, setSimMessage] = useState(null);

  // Pomodoro Timer State
  const [timerMode, setTimerMode] = useState("focus");
  const [initialMinutes, setInitialMinutes] = useState(25);
  const [timerSeconds, setTimerSeconds] = useState(25 * 60);
  const [isTimerRunning, setIsTimerRunning] = useState(false);
  const [selectedPomoCourse, setSelectedPomoCourse] = useState("Object Oriented Programming");

  // Flashcards state
  const [currentFcIndex, setCurrentFcIndex] = useState(0);
  const [isFlipped, setIsFlipped] = useState(false);

  // Modals
  const [modalTaskOpen, setModalTaskOpen] = useState(false);
```

```
  const [modalCourseOpen, setModalCourseOpen] = useState(false);
  const [modalFcOpen, setModalFcOpen] = useState(false);

  // Form states
  const [newTask, setNewTask] = useState({ title: "", course: "", dueDate: "", priority: "Medium", hours: 3 });
  const [newCourse, setNewCourse] = useState({ code: "", name: "", credits: "4.0", grade: "A" });
  const [newFc, setNewFc] = useState({ subject: "", question: "", answer: "" });

  // Compute SGPA dynamically
  const { sgpa, totalCredits, totalWeighted } = useMemo(() => {
    let creds = 0;
    let weighted = 0;
    courses.forEach(c => {
      const pt = GRADE_POINTS[c.grade] ?? c.gradePoint ?? 0.0;
      creds += c.credits;
      weighted += (c.credits * pt);
    });
    return {
      sgpa: creds > 0 ? (weighted / creds) : 0.0,
      totalCredits: creds,
      totalWeighted: weighted
    };
  }, [courses]);

  // Pomodoro interval ticker
  useEffect(() => {
    let interval = null;
    if (isTimerRunning) {
      interval = setInterval(() => {
        setTimerSeconds(prev => {
          if (prev > 1) return prev - 1;
          clearInterval(interval);
          handleTimerComplete();
          return 0;
        });
      }, 1000);
    }
    return () => clearInterval(interval);
  }, [isTimerRunning, timerMode, selectedPomoCourse]);

  const handleTimerComplete = async () => {
    setIsTimerRunning(false);
    if (timerMode === "focus") {
      try {
        await fetch("/api/pomodoro", {
```

```
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ subject: selectedPomoCourse, durationMinutes: initialMinutes })
      });
      setPomodoroStats(prev => ({
        completedSessions: prev.completedSessions + 1,
        totalMinutes: prev.totalMinutes + initialMinutes,
        streakDays: prev.streakDays + 1
      }));
    } catch (err) {
      console.error("Failed to log pomodoro to database:", err);
    }
    alert("Focus interval completed. Take a five-minute rest.");
  } else {
    alert("Break completed. Ready for the next focus sprint?");
  }
  setTimerSeconds(initialMinutes * 60);
};

const setPomoMode = (mode, mins) => {
  setIsTimerRunning(false);
  setTimerMode(mode);
  setInitialMinutes(mins);
  setTimerSeconds(mins * 60);
};

// Keyboard shortcut for flashcards (Spacebar to flip)
useEffect(() => {
  const handleKeyDown = (e) => {
    if (e.code === "Space" && activeTab === "flashcards" && !modalTaskOpen && !modalCourseOpen && !modalFcOpen) {
      e.preventDefault();
      setIsFlipped(prev => !prev);
    }
  };
  window.addEventListener("keydown", handleKeyDown);
  return () => window.removeEventListener("keydown", handleKeyDown);
}, [activeTab, modalTaskOpen, modalCourseOpen, modalFcOpen]);

// Task Handlers
const handleToggleTask = async (id, currentCompleted) => {
  setTasks(prev => prev.map(t => t.id === id ? { ...t, completed: !currentCompleted } : t));
  try {
    await fetch("/api/tasks", {
      method: "PATCH",
      headers: { "Content-Type": "application/json" },
```

```
      body: JSON.stringify({ id, completed: !currentCompleted })
    });
  } catch (e) {
    console.error("Task toggle error:", e);
  }
};

const handleDeleteTask = async (id) => {
  setTasks(prev => prev.filter(t => t.id !== id));
  try {
    await fetch(`/api/tasks?id=${encodeURIComponent(id)}`, { method: "DELETE" });
  } catch (e) {
    console.error("Task delete error:", e);
  }
};

const handleCreateTask = async (e) => {
  e.preventDefault();
  const createdId = "T-" + Date.now();
  const taskObj = { ...newTask, id: createdId, hours: Number(newTask.hours), completed: false };
  setTasks(prev => [taskObj, ...prev]);
  setModalTaskOpen(false);
  setNewTask({ title: "", course: "", dueDate: "", priority: "Medium", hours: 3 });

  try {
    await fetch("/api/tasks", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify(taskObj)
    });
  } catch (e) {
    console.error("Task create error:", e);
  }
};

const handleSortTasks = () => {
  setTasks(prev => [...prev].sort((a, b) => {
    const pMap = { High: 1, Medium: 2, Low: 3 };
    if (pMap[a.priority] !== pMap[b.priority]) return pMap[a.priority] - pMap[b.priority];
    return new Date(a.dueDate) - new Date(b.dueDate);
  }));
};

// Course Handlers
const handleCreateCourse = async (e) => {
```

```
    e.preventDefault();
    const createdId = "C-" + Date.now();
    const pt = GRADE_POINTS[newCourse.grade] || 0.0;
    const courseObj = {
      id: createdId,
      code: newCourse.code.trim(),
      name: newCourse.name.trim(),
      credits: parseFloat(newCourse.credits),
      grade: newCourse.grade,
      gradePoint: pt
    };
    setCourses(prev => [...prev, courseObj]);
    setModalCourseOpen(false);
    setNewCourse({ code: "", name: "", credits: "4.0", grade: "A" });

    try {
      await fetch("/api/courses", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(courseObj)
      });
    } catch (e) {
      console.error("Course create error:", e);
    }
  };

  const handleDeleteCourse = async (id) => {
    setCourses(prev => prev.filter(c => c.id !== id));
    try {
      await fetch(`/api/courses?id=${encodeURIComponent(id)}`, { method: "DELETE" });
    } catch (e) {
      console.error("Course delete error:", e);
    }
  };

  // Flashcard Handlers
  const handleToggleMastered = async () => {
    if (flashcards.length === 0) return;
    const active = flashcards[currentFcIndex];
    const newMastered = !active.mastered;
    setFlashcards(prev => prev.map((f, i) => i === currentFcIndex ? { ...f, mastered: newMastered } : f));

    try {
      await fetch("/api/flashcards", {
        method: "PATCH",
```

```
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ id: active.id, mastered: newMastered })
      });
    } catch (e) {
      console.error("Flashcard toggle error:", e);
    }
  };

  const handleCreateFlashcard = async (e) => {
    e.preventDefault();
    const cardObj = {
      id: "FC-" + Date.now(),
      subject: newFc.subject.trim(),
      question: newFc.question.trim(),
      answer: newFc.answer.trim(),
      mastered: false
    };
    setFlashcards(prev => [...prev, cardObj]);
    setCurrentFcIndex(flashcards.length);
    setIsFlipped(false);
    setModalFcOpen(false);
    setNewFc({ subject: "", question: "", answer: "" });

    try {
      await fetch("/api/flashcards", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(cardObj)
      });
    } catch (e) {
      console.error("Flashcard create error:", e);
    }
  };

  // GPA Target simulation
  const handleCalculateTarget = () => {
    const target = parseFloat(simTarget);
    if (!target || totalCredits === 0) return;
    const diff = target - sgpa;
    if (diff <= 0) {
      setSimMessage({
        type: "success",
        text: `Target achieved. Current SGPA (${sgpa.toFixed(2)}) meets or exceeds target (${target.toFixed(2)}).`
      });
    } else {
```

```
      const neededPoints = (target * (totalCredits + 4)) - totalWeighted;
      const gradeReq = (neededPoints / 4).toFixed(1);
      setSimMessage({
        type: "info",
        text: `To achieve ${target.toFixed(2)}, an average of ${gradeReq} / 10.0 grade points is required on upcoming
 modules.`
      });
    }
  };

  // Downloads
  const downloadTextReport = () => {
    const text = `================================================================================
                  STUDENT WORKSPACE • ACADEMIC REPORT
================================================================================
Generated Date:   ${new Date().toISOString().split('T')[0]}
Workspace:        Student Workspace
Current SGPA:     ${sgpa.toFixed(2)} / 10.00
Enrolled Credits: ${totalCredits.toFixed(1)} Credits
Study Time:       ${pomodoroStats.totalMinutes} minutes (${pomodoroStats.completedSessions} focus blocks)
Local Database:   SQLite WebAssembly Engine
================================================================================

1. REGISTERED SEMESTER COURSES:
--------------------------------------------------------------------------------
Code          Course Title                              Credits   Grade   Points
--------------------------------------------------------------------------------
${courses.length === 0 ? "No courses recorded yet." : courses.map(c => `${c.code.padEnd(13)} ${c.name.padEnd(41)} ${c
.credits.toFixed(1).padEnd(9)} ${c.grade.padEnd(7)} ${(GRADE_POINTS[c.grade] || 0).toFixed(1)}`).join('\n')}
--------------------------------------------------------------------------------

2. ASSIGNMENTS & DELIVERABLES:
--------------------------------------------------------------------------------
Title                                   Course                  Due Date     Priority
--------------------------------------------------------------------------------
${tasks.length === 0 ? "No assignments recorded yet." : tasks.map(t => `${(t.completed ? '[DONE]' : '[DUE] ').padEnd(
7)} ${t.title.padEnd(35)} ${t.course.padEnd(23)} ${t.dueDate.padEnd(12)} ${t.priority}`).join('\n')}
--------------------------------------------------------------------------------

3. FLASHCARDS RETENTION:
Total Cards:      ${flashcards.length}
Mastered Topics:  ${flashcards.filter(f => f.mastered).length}
================================================================================`;

    const blob = new Blob([text], { type: "text/plain" });
    const url = URL.createObjectURL(blob);
    const a = document.createElement("a");
    a.href = url;
```

```
    a.download = "Student_Workspace_Academic_Report.txt";
    a.click();
    URL.revokeObjectURL(url);
  };

  const downloadJsonBackup = () => {
    const payload = JSON.stringify({ courses, tasks, flashcards, pomodoroStats }, null, 2);
    const blob = new Blob([payload], { type: "application/json" });
    const url = URL.createObjectURL(blob);
    const a = document.createElement("a");
    a.href = url;
    a.download = "student_suite_backup.json";
    a.click();
    URL.revokeObjectURL(url);
  };

  // Filtered tasks
  const filteredTasks = useMemo(() => {
    const q = taskSearch.toLowerCase();
    return tasks.filter(t => {
      const matchQ = t.title.toLowerCase().includes(q) || t.course.toLowerCase().includes(q);
      const matchP = priorityFilter === "ALL" || t.priority === priorityFilter;
      const matchS = statusFilter === "ALL" || (statusFilter === "COMPLETED" ? t.completed : !t.completed);
      return matchQ && matchP && matchS;
    });
  }, [tasks, taskSearch, priorityFilter, statusFilter]);

  // Pomodoro SVG calculations
  const totalCircumference = 2 * Math.PI * 104; // r = 104 -> ~653.45
  const timerFraction = timerSeconds / (initialMinutes * 60);
  const ringOffset = totalCircumference * (1 - timerFraction);

  const formatTimerDigits = () => {
    const m = Math.floor(timerSeconds / 60);
    const s = timerSeconds % 60;
    return `${String(m).padStart(2, "0")}:${String(s).padStart(2, "0")}`;
  };

  const pendingTaskCount = tasks.filter(t => !t.completed).length;
  const currentCard = flashcards[currentFcIndex] || null;

  return (
    <>
      <Head>
        <title>Student Workspace</title>
```

```
        <meta name="description" content="A calm, offline-first academic productivity workspace powered by local SQLi
te." />
      </Head>

      <div className="app-layout">
        {/* Top Header */}
        <header className="app-header">
          <div className="header-left">
            <a href="#assignments" className="brand-link" onClick={() => setActiveTab("assignments")}>
              <span className="brand-mark">
                <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" str
okeLinecap="round" strokeLinejoin="round">
                  <path d="M12 2v20M17 5H9.5a3.5 3.5 0 0 0 0 7h5a3.5 3.5 0 0 1 0 7H6"/>
                </svg>
              </span>
              <span className="brand-name">Student Workspace</span>
            </a>
            <span className="header-divider">/</span>
            <span className="header-context">Academic Workspace</span>
          </div>

          <div className="header-center">
            <div className="student-chip">
              <span className="student-avatar">SW</span>
              <span className="student-name">Student Workspace</span>
              <span className="student-meta">Academic Workspace</span>
            </div>
          </div>

          <div className="header-right">
            <div className="status-metric">
              <span className="metric-label">SGPA</span>
              <span className="metric-value">{sgpa.toFixed(2)}</span>
            </div>
            <div className="status-divider"></div>
            <div className="status-metric">
              <span className="metric-label">Due Soon</span>
              <span className="metric-value">{pendingTaskCount}</span>
            </div>
            <div className="status-divider"></div>
            <div className="status-metric">
              <span className="metric-label">Focus Blocks</span>
              <span className="metric-value">{pomodoroStats.completedSessions}</span>
            </div>
          </div>
        </header>
```

```
        {/* Navigation Tabs */}
        <nav className="app-nav">
          <div className="nav-container">
            <div className="nav-tabs">
              <button className={`nav-tab ${activeTab === "assignments" ? "active" : ""}`} onClick={() => setActiveTa
b("assignments")}>
                <svg className="tab-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor
" strokeWidth="1.5" strokeLinecap="round" strokeLinejoin="round">
                  <path d="M9 11l3 3L22 4"/><path d="M21 12v7a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h11"/>
                </svg>
                <span>Assignments</span>
              </button>
              <button className={`nav-tab ${activeTab === "gpa" ? "active" : ""}`} onClick={() => setActiveTab("gpa")
}>
                <svg className="tab-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor
" strokeWidth="1.5" strokeLinecap="round" strokeLinejoin="round">
                  <path d="M22 10v6M2 10l10-5 10 5-10 5z"/><path d="M6 12v5c3 3 9 3 12 0v-5"/>
                </svg>
                <span>SGPA Calculator</span>
              </button>
              <button className={`nav-tab ${activeTab === "pomodoro" ? "active" : ""}`} onClick={() => setActiveTab("
pomodoro")}>
                <svg className="tab-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor
" strokeWidth="1.5" strokeLinecap="round" strokeLinejoin="round">
                  <circle cx="12" cy="12" r="10"/><polyline points="12 6 12 12 16 14"/>
                </svg>
                <span>Focus Timer</span>
              </button>
              <button className={`nav-tab ${activeTab === "flashcards" ? "active" : ""}`} onClick={() => setActiveTab
("flashcards")}>
                <svg className="tab-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor
" strokeWidth="1.5" strokeLinecap="round" strokeLinejoin="round">
                  <rect x="2" y="7" width="20" height="14" rx="2" ry="2"/><path d="M16 3H4a2 2 0 0 0-2 2v2"/>
                </svg>
                <span>Revision Flashcards</span>
              </button>
              <button className={`nav-tab ${activeTab === "analytics" ? "active" : ""}`} onClick={() => setActiveTab(
"analytics")}>
                <svg className="tab-icon" width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor
" strokeWidth="1.5" strokeLinecap="round" strokeLinejoin="round">
                  <line x1="18" y1="20" x2="18" y2="10"/><line x1="12" y1="20" x2="12" y2="4"/><line x1="6" y1="20" x
2="6" y2="14"/>
                </svg>
                <span>Summary & Export</span>
              </button>
            </div>
          </div>
        </nav>

        {/* Main Content Area */}
        <main className="app-main">

          {/* ==================== TAB 1: ASSIGNMENTS ==================== */}
          {activeTab === "assignments" && (
            <section className="tab-pane active">
              <div className="section-header">
```

```
                <div>
                  <h1 className="section-title">Assignments and deadlines</h1>
                  <p className="section-desc">Track and prioritize coursework with due dates and estimated workload.<
/p>
                </div>
                <div className="section-actions">
                  <button className="btn btn-secondary" onClick={handleSortTasks}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><path d="M7 15l5 5 5-5M7 9l5-5 5 5"/></svg>
                    <span>Sort by priority</span>
                  </button>
                  <button className="btn btn-primary" onClick={() => setModalTaskOpen(true)}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><line x1="12" y1="5" x2="12" y2="19"/><line x1="5" y1="12" x2="19" y2="12"/></svg>
                    <span>New assignment</span>
                  </button>
                </div>
              </div>

              {/* Filter Panel */}
              <div className="filter-panel">
                <div className="search-field">
                  <svg className="field-icon" width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentC
olor" strokeWidth="1.5"><circle cx="11" cy="11" r="8"/><line x1="21" y1="21" x2="16.65" y2="16.65"/></svg>
                  <input
                    type="text"
                    value={taskSearch}
                    onChange={(e) => setTaskSearch(e.target.value)}
                    placeholder="Search assignments or courses"
                  />
                </div>
                <div className="filter-controls">
                  <div className="select-wrapper">
                    <label className="select-label">Priority</label>
                    <select value={priorityFilter} onChange={(e) => setPriorityFilter(e.target.value)} className="for
m-select-sm">
                      <option value="ALL">All priorities</option>
                      <option value="High">High</option>
                      <option value="Medium">Medium</option>
                      <option value="Low">Low</option>
                    </select>
                  </div>
                  <div className="select-wrapper">
                    <label className="select-label">Status</label>
                    <select value={statusFilter} onChange={(e) => setStatusFilter(e.target.value)} className="form-se
lect-sm">
                      <option value="ALL">All status</option>
                      <option value="PENDING">Pending</option>
                      <option value="COMPLETED">Completed</option>
                    </select>
                  </div>
```

```
              </div>
            </div>

            {/* Tasks Grid */}
            <div className="task-grid">
              {filteredTasks.length === 0 ? (
                <div style={{ gridColumn: "1/-1", padding: "32px", textAlign: "center", color: "var(--text-muted)",
 background: "white", borderRadius: "var(--radius-card)", border: "1px dashed var(--border-hairline)" }}>
                  <p>No assignments found matching current filter criteria.</p>
                </div>
              ) : (
                filteredTasks.map(t => (
                  <div key={t.id} className={`task-card ${t.completed ? "completed" : ""}`}>
                    <div className="task-card-header">
                      <div>
                        <h2 className="task-title">{t.title}</h2>
                        <p className="task-course">{t.course}</p>
                      </div>
                      <span className={`priority-tag priority-${t.priority.toLowerCase()}`}>{t.priority}</span>
                    </div>
                    <div className="task-meta-row">
                      <span>Due {t.dueDate}</span>
                      <span>{t.hours} hrs est.</span>
                    </div>
                    <div className="task-footer-actions">
                      <button className="btn btn-sm btn-secondary" onClick={() => handleToggleTask(t.id, t.complete
d)}>
                        {t.completed ? "Reopen" : "Mark completed"}
                      </button>
                      <button className="btn btn-sm btn-ghost" onClick={() => handleDeleteTask(t.id)}>
                        Delete
                      </button>
                    </div>
                  </div>
                ))
              )}
            </div>
          </section>
        )}

        {/* ==================== TAB 2: GPA CALCULATOR ==================== */}
        {activeTab === "gpa" && (
          <section className="tab-pane active">
            <div className="section-header">
              <div>
                <h1 className="section-title">SGPA and grade calculation</h1>
                <p className="section-desc">Weighted credit point calculations based on the standard 10-point scale
.</p>
```

```
                </div>
                <button className="btn btn-primary" onClick={() => setModalCourseOpen(true)}>
                  <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.5"
><line x1="12" y1="5" x2="12" y2="19"/><line x1="5" y1="12" x2="19" y2="12"/></svg>
                  <span>Add course</span>
                </button>
              </div>

              <div className="gpa-overview-grid">
                <div className="card gpa-metric-card">
                  <div className="metric-top">
                    <span className="card-caption">Semester SGPA</span>
                    <span className="status-tag tag-green">
                      {sgpa >= 9.0 ? "First class with distinction" : sgpa >= 8.0 ? "First class with honors" : sgpa
>= 6.5 ? "First class" : "Passing standing"}
                    </span>
                  </div>
                  <div className="gpa-display-row">
                    <span className="gpa-large-number">{sgpa.toFixed(2)}</span>
                    <span className="gpa-max-scale">/ 10.00</span>
                  </div>
                  <div className="gpa-breakdown-row">
                    <div className="breakdown-item">
                      <span className="breakdown-label">Registered credits</span>
                      <span className="breakdown-val">{totalCredits.toFixed(1)}</span>
                    </div>
                    <div className="breakdown-item">
                      <span className="breakdown-label">Weighted points</span>
                      <span className="breakdown-val">{totalWeighted.toFixed(1)}</span>
                    </div>
                  </div>
                </div>

                <div className="card gpa-simulator-card">
                  <div className="metric-top">
                    <span className="card-caption">Target SGPA simulator</span>
                    <span className="status-tag tag-neutral">Forecasting</span>
                  </div>
                  <p className="simulator-intro">Estimate required grade points across remaining semester modules to
reach your target.</p>
                  <div className="sim-controls">
                    <div className="input-inline">
                      <label htmlFor="simTargetInput">Target CGPA</label>
                      <input
                        id="simTargetInput"
                        type="number"
                        step="0.05"
                        min="5.0"
```

```
                        max="10.0"
                        value={simTarget}
                        onChange={(e) => setSimTarget(e.target.value)}
                      />
                    </div>
                    <button className="btn btn-secondary" onClick={handleCalculateTarget}>Calculate requirement</butt
on>
                  </div>
                  <div className="sim-result-box">
                    {simMessage ? simMessage.text : "Enter target grade above to view credit requirement analysis."}
                  </div>
                </div>
              </div>

              {/* Courses Table */}
              <div className="card table-card">
                <table className="data-table">
                  <thead>
                    <tr>
                      <th style={{ width: "140px" }}>Course code</th>
                      <th>Course title</th>
                      <th style={{ width: "90px", textAlign: "right" }}>Credits</th>
                      <th style={{ width: "90px", textAlign: "center" }}>Grade</th>
                      <th style={{ width: "110px", textAlign: "right" }}>Points</th>
                      <th style={{ width: "130px", textAlign: "right" }}>Weighted pts</th>
                      <th style={{ width: "80px", textAlign: "right" }}>Action</th>
                    </tr>
                  </thead>
                  <tbody>
                    {courses.map(c => {
                      const pt = GRADE_POINTS[c.grade] ?? 0.0;
                      return (
                        <tr key={c.id}>
                          <td><span className="code-pill">{c.code}</span></td>
                          <td>{c.name}</td>
                          <td style={{ textAlign: "right" }}>{c.credits.toFixed(1)}</td>
                          <td style={{ textAlign: "center" }}><span className="status-tag tag-neutral">{c.grade}</spa
n></td>
                          <td style={{ textAlign: "right" }}>{pt.toFixed(1)}</td>
                          <td style={{ textAlign: "right" }}><strong>{(c.credits * pt).toFixed(1)}</strong></td>
                          <td style={{ textAlign: "right" }}>
                            <button className="btn btn-sm btn-ghost" onClick={() => handleDeleteCourse(c.id)}>Remove<
/button>
                          </td>
                        </tr>
                      );
                    })}
                  </tbody>
```

```
                </table>
              </div>
            </section>
          )}

          {/* ==================== TAB 3: FOCUS TIMER ==================== */}
          {activeTab === "pomodoro" && (
            <section className="tab-pane active">
              <div className="section-header">
                <div>
                  <h1 className="section-title">Focus timer</h1>
                  <p className="section-desc">Structured 25-minute study intervals to sustain deliberate attention.</
p>
                </div>
              </div>

              <div className="pomodoro-grid">
                <div className="card timer-card">
                  <div className="mode-selector">
                    <button className={`mode-btn ${timerMode === "focus" ? "active" : ""}`} onClick={() => setPomoMod
e("focus", 25)}>Focus (25m)</button>
                    <button className={`mode-btn ${timerMode === "shortBreak" ? "active" : ""}`} onClick={() => setPo
moMode("shortBreak", 5)}>Short break (5m)</button>
                    <button className={`mode-btn ${timerMode === "longBreak" ? "active" : ""}`} onClick={() => setPom
oMode("longBreak", 15)}>Long break (15m)</button>
                  </div>

                  <div className="dial-container">
                    <svg className="dial-svg" width="240" height="240" viewBox="0 0 240 240">
                      <circle className="dial-track" cx="120" cy="120" r="104" />
                      <circle
                        className="dial-indicator"
                        cx="120"
                        cy="120"
                        r="104"
                        style={{ strokeDashoffset: ringOffset }}
                      />
                    </svg>
                    <div className="dial-center">
                      <span className="timer-readout">{formatTimerDigits()}</span>
                      <span className="timer-status-text">{timerMode === "focus" ? "Focus interval" : "Rest interval"
}</span>
                    </div>
                  </div>

                  <div className="timer-action-row">
                    {!isTimerRunning ? (
                      <button className="btn btn-primary btn-md" onClick={() => setIsTimerRunning(true)}>
                        <svg width="14" height="14" viewBox="0 0 24 24" fill="currentColor"><polygon points="5 3 19 1
2 5 21 5 3"/></svg>
                        <span>Start</span>
```

```
                    </button>
                  ) : (
                    <button className="btn btn-secondary btn-md" onClick={() => setIsTimerRunning(false)}>
                      <svg width="14" height="14" viewBox="0 0 24 24" fill="currentColor"><rect x="6" y="4" width="
4" height="16"/><rect x="14" y="4" width="4" height="16"/></svg>
                      <span>Pause</span>
                    </button>
                  )}
                  <button className="btn btn-secondary btn-md" onClick={() => { setIsTimerRunning(false); setTimerS
econds(initialMinutes * 60); }}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="
1.5"><path d="M3 12a9 9 0 1 0 9-9 9.75 9.75 0 0 0-6.74 2.74L3 8"/><path d="M3 3v5h5"/></svg>
                    <span>Reset</span>
                  </button>
                </div>

                <div className="timer-meta-strip">
                  <div className="meta-stat">
                    <span className="meta-num">{pomodoroStats.completedSessions}</span>
                    <span className="meta-lbl">Completed blocks</span>
                  </div>
                  <div className="meta-stat-divider"></div>
                  <div className="meta-stat">
                    <span className="meta-num">{pomodoroStats.totalMinutes}m</span>
                    <span className="meta-lbl">Total focus</span>
                  </div>
                  <div className="meta-stat-divider"></div>
                  <div className="meta-stat">
                    <span className="meta-num">{pomodoroStats.streakDays}</span>
                    <span className="meta-lbl">Session streak</span>
                  </div>
                </div>
              </div>

              <div className="timer-side-column">
                <div className="card">
                  <h2 className="card-title">Study session protocol</h2>
                  <ul className="guideline-list">
                    <li>
                      <span className="step-num">1</span>
                      <div>
                        <strong>Select one deliverable</strong>
                        <p>Silence notifications and work only on the selected subject.</p>
                      </div>
                    </li>
                    <li>
                      <span className="step-num">2</span>
                      <div>
```

```
                        <strong>Maintain continuous focus</strong>
                        <p>Avoid tab switching or passive browsing until the timer ends.</p>
                      </div>
                    </li>
                    <li>
                      <span className="step-num">3</span>
                      <div>
                        <strong>Take deliberate rest</strong>
                        <p>Step away from the screen for five minutes to recover attention.</p>
                      </div>
                    </li>
                  </ul>
                </div>

                <div className="card">
                  <h2 className="card-title">Active course focus</h2>
                  <div className="form-group" style={{ marginBottom: 0 }}>
                    <label htmlFor="pomoSubjectSelector">Log session under course</label>
                    <select
                      id="pomoSubjectSelector"
                      className="form-select"
                      value={selectedPomoCourse}
                      onChange={(e) => setSelectedPomoCourse(e.target.value)}
                    >
                      {courses.map(c => (
                        <option key={c.id} value={c.name}>{c.name}</option>
                      ))}
                    </select>
                  </div>
                </div>
              </div>
            </section>
          )}

          {/* ==================== TAB 4: FLASHCARDS ==================== */}
          {activeTab === "flashcards" && (
            <section className="tab-pane active">
              <div className="section-header">
                <div>
                  <h1 className="section-title">Revision flashcards</h1>
                  <p className="section-desc">Active recall practice to strengthen concept retrieval before examinati
ons.</p>
                </div>
                <button className="btn btn-primary" onClick={() => setModalFcOpen(true)}>
                  <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.5"
><line x1="12" y1="5" x2="12" y2="19"/><line x1="5" y1="12" x2="19" y2="12"/></svg>
```

```
                  <span>New card</span>
                </button>
              </div>

              <div className="flashcard-deck-layout">
                {currentCard ? (
                  <div className="flashcard-viewport" onClick={() => setIsFlipped(prev => !prev)}>
                    <div className={`flashcard-box ${isFlipped ? "flipped" : ""}`}>
                      <div className="card-side side-front">
                        <div className="side-header">
                          <span className="tag-category">{currentCard.subject}</span>
                          <span className="side-counter">Card {currentFcIndex + 1} of {flashcards.length}</span>
                        </div>
                        <div className="side-content">
                          <span className="content-eyebrow">Prompt</span>
                          <p className="content-text">{currentCard.question}</p>
                        </div>
                        <div className="side-footer">
                          <span>Press <kbd className="keycap">Space</kbd> or click to reveal answer</span>
                        </div>
                      </div>

                      <div className="card-side side-back">
                        <div className="side-header">
                          <span className="tag-category">Answer explanation</span>
                          <span className="mastery-indicator">{currentCard.mastered ? "Mastered" : "Unmastered"}</spa
n>
                        </div>
                        <div className="side-content">
                          <span className="content-eyebrow">Answer</span>
                          <p className="content-text">{currentCard.answer}</p>
                        </div>
                        <div className="side-footer">
                          <span>Click to flip back to question</span>
                        </div>
                      </div>
                    </div>
                  </div>
                ) : (
                  <div className="card" style={{ padding: "40px", textAlign: "center" }}>
                    <p>No revision cards available. Create one to begin active recall.</p>
                  </div>
                )}

                <div className="flashcard-nav-row">
                  <button className="btn btn-secondary" onClick={() => { setIsFlipped(false); setCurrentFcIndex(prev
=> (prev - 1 + flashcards.length) % flashcards.length); }}>
```

```
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><polyline points="15 18 9 12 15 6"/></svg>
                    <span>Previous</span>
                  </button>
                  <button className="btn btn-secondary" onClick={() => setIsFlipped(prev => !prev)}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><path d="M21.5 2v6h-6M2.5 22v-6h6M2 11.5a10 10 0 0 1 18.8-4.3M22 12.5a10 10 0 0 1-18.8 4.2"/></svg>
                    <span>Flip card</span>
                  </button>
                  <button className="btn btn-secondary" onClick={handleToggleMastered}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><polygon points="12 2 15.09 8.26 22 9.27 17 14.14 18.18 21.02 12 17.77 5.82 21.02 7 14.14 2 9.27 8.91 8.26 12 2"/>
</svg>
                    <span>{currentCard?.mastered ? "Mastered" : "Mark mastered"}</span>
                  </button>
                  <button className="btn btn-secondary" onClick={() => { setIsFlipped(false); setCurrentFcIndex(prev
=> (prev + 1) % flashcards.length); }}>
                    <span>Next</span>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><polyline points="9 18 15 12 9 6"/></svg>
                  </button>
                </div>
              </div>
            </section>
          )}

          {/* ==================== TAB 5: ANALYTICS ==================== */}
          {activeTab === "analytics" && (
            <section className="tab-pane active">
              <div className="section-header">
                <div>
                  <h1 className="section-title">Academic report and export</h1>
                  <p className="section-desc">Generate official performance documentation or download local state bac
kups.</p>
                </div>
                <div className="section-actions">
                  <button className="btn btn-secondary" onClick={downloadTextReport}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    <span>Download text report</span>
                  </button>
                  <button className="btn btn-primary" onClick={downloadJsonBackup}>
                    <svg width="14" height="14" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="1.
5"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2
="12" y2="3"/></svg>
                    <span>Download JSON backup</span>
                  </button>
                </div>
              </div>

              <div className="analytics-stat-row">
                <div className="card stat-card">
                  <span className="stat-caption">Registered courses</span>
                  <span className="stat-metric-large">{courses.length}</span>
                </div>
```

```
                <div className="card stat-card">
                  <span className="stat-caption">Logged assignments</span>
                  <span className="stat-metric-large">{tasks.length}</span>
                </div>
                <div className="card stat-card">
                  <span className="stat-caption">Mastered concepts</span>
                  <span className="stat-metric-large">{flashcards.filter(f => f.mastered).length} / {flashcards.lengt
h}</span>
                </div>
                <div className="card stat-card">
                  <span className="stat-caption">Total focus logged</span>
                  <span className="stat-metric-large">{pomodoroStats.totalMinutes}m</span>
                </div>
              </div>

              <div className="card report-box">
                <div className="report-box-header">
                  <span className="card-title">Generated summary preview</span>
                  <span className="status-tag tag-neutral">SQLite Database Engine</span>
                </div>
                <pre className="ascii-preview">
{`================================================================================
                    STUDENT WORKSPACE • ACADEMIC REPORT
================================================================================
Workspace:        Student Workspace
Current SGPA:     ${sgpa.toFixed(2)} / 10.00
Enrolled Credits: ${totalCredits.toFixed(1)} Credits
Study Time:       ${pomodoroStats.totalMinutes} minutes (${pomodoroStats.completedSessions} focus blocks)
Database:         Embedded SQLite WebAssembly Engine
================================================================================`}
                </pre>
              </div>
            </section>
          )}

        </main>

        {/* Footer */}
        <footer className="app-footer">
          <div className="footer-container">
            <span className="footer-left">Student Workspace</span>
            <span className="footer-center">Offline-First Academic Productivity • SQLite WebAssembly Engine</span>
            <span className="footer-right">Local Persistence</span>
          </div>
        </footer>
      </div>
```

```
      {/* Modal 1: Assignment */}
      {modalTaskOpen && (
        <div className="modal-backdrop active">
          <div className="modal-dialog">
            <div className="modal-header">
              <h2 className="modal-title">New assignment</h2>
              <button className="btn-close" onClick={() => setModalTaskOpen(false)}>X</button>
            </div>
            <form onSubmit={handleCreateTask}>
              <div className="form-group">
                <label>Assignment title</label>
                <input
                  type="text"
                  required
                  value={newTask.title}
                  onChange={(e) => setNewTask({ ...newTask, title: e.target.value })}
                  placeholder="Multithreaded Producer-Consumer Concurrency"
                />
              </div>
              <div className="form-row-2">
                <div className="form-group">
                  <label>Course or module</label>
                  <input
                    type="text"
                    required
                    value={newTask.course}
                    onChange={(e) => setNewTask({ ...newTask, course: e.target.value })}
                    placeholder="Object Oriented Programming"
                  />
                </div>
                <div className="form-group">
                  <label>Due date</label>
                  <input
                    type="date"
                    required
                    value={newTask.dueDate}
                    onChange={(e) => setNewTask({ ...newTask, dueDate: e.target.value })}
                  />
                </div>
              </div>
              <div className="form-row-2">
                <div className="form-group">
                  <label>Priority</label>
                  <select
```

```
                    value={newTask.priority}
                    onChange={(e) => setNewTask({ ...newTask, priority: e.target.value })}
                  >
                    <option value="High">High</option>
                    <option value="Medium">Medium</option>
                    <option value="Low">Low</option>
                  </select>
                </div>
                <div className="form-group">
                  <label>Estimated hours</label>
                  <input
                    type="number"
                    min="1"
                    max="50"
                    value={newTask.hours}
                    onChange={(e) => setNewTask({ ...newTask, hours: e.target.value })}
                  />
                </div>
              </div>
              <div className="modal-actions">
                <button type="button" className="btn btn-secondary" onClick={() => setModalTaskOpen(false)}>Cancel</b
utton>
                <button type="submit" className="btn btn-primary">Save assignment</button>
              </div>
            </form>
          </div>
        </div>
      )}

      {/* Modal 2: Course */}
      {modalCourseOpen && (
        <div className="modal-backdrop active">
          <div className="modal-dialog">
            <div className="modal-header">
              <h2 className="modal-title">Add semester course</h2>
              <button className="btn-close" onClick={() => setModalCourseOpen(false)}>✕</button>
            </div>
            <form onSubmit={handleCreateCourse}>
              <div className="form-row-2">
                <div className="form-group">
                  <label>Course code</label>
                  <input
                    type="text"
                    required
                    value={newCourse.code}
                    onChange={(e) => setNewCourse({ ...newCourse, code: e.target.value })}
```

```
                placeholder="25CSH-208"
              />
            </div>
            <div className="form-group">
              <label>Course title</label>
              <input
                type="text"
                required
                value={newCourse.name}
                onChange={(e) => setNewCourse({ ...newCourse, name: e.target.value })}
                placeholder="Object Oriented Programming"
              />
            </div>
          </div>
          <div className="form-row-2">
            <div className="form-group">
              <label>Credits</label>
              <select
                value={newCourse.credits}
                onChange={(e) => setNewCourse({ ...newCourse, credits: e.target.value })}
              >
                <option value="1.0">1.0</option>
                <option value="2.0">2.0</option>
                <option value="3.0">3.0</option>
                <option value="3.5">3.5</option>
                <option value="4.0">4.0</option>
                <option value="5.0">5.0</option>
              </select>
            </div>
            <div className="form-group">
              <label>Expected grade</label>
              <select
                value={newCourse.grade}
                onChange={(e) => setNewCourse({ ...newCourse, grade: e.target.value })}
              >
                <option value="A+">A+ (10.0)</option>
                <option value="A">A (9.0)</option>
                <option value="B+">B+ (8.0)</option>
                <option value="B">B (7.0)</option>
                <option value="C+">C+ (6.0)</option>
                <option value="C">C (5.0)</option>
                <option value="D">D (4.0)</option>
                <option value="F">F (0.0)</option>
              </select>
            </div>
```

```
              </div>
              <div className="modal-actions">
                <button type="button" className="btn btn-secondary" onClick={() => setModalCourseOpen(false)}>Cancel<
/button>
                <button type="submit" className="btn btn-primary">Add course</button>
              </div>
            </form>
          </div>
        </div>
      )}

      {/* Modal 3: Flashcard */}
      {modalFcOpen && (
        <div className="modal-backdrop active">
          <div className="modal-dialog">
            <div className="modal-header">
              <h2 className="modal-title">Create revision card</h2>
              <button className="btn-close" onClick={() => setModalFcOpen(false)}>✕</button>
            </div>
            <form onSubmit={handleCreateFlashcard}>
              <div className="form-group">
                <label>Topic</label>
                <input
                  type="text"
                  required
                  value={newFc.subject}
                  onChange={(e) => setNewFc({ ...newFc, subject: e.target.value })}
                  placeholder="Java OOP • Polymorphism"
                />
              </div>
              <div className="form-group">
                <label>Question prompt</label>
                <textarea
                  rows="3"
                  required
                  value={newFc.question}
                  onChange={(e) => setNewFc({ ...newFc, question: e.target.value })}
                  placeholder="Enter concept prompt..."
                />
              </div>
              <div className="form-group">
                <label>Answer</label>
                <textarea
                  rows="4"
                  required
                  value={newFc.answer}
```

```
                  onChange={(e) => setNewFc({ ...newFc, answer: e.target.value })}
                  placeholder="Enter concise explanation..."
                />
              </div>
              <div className="modal-actions">
                <button type="button" className="btn btn-secondary" onClick={() => setModalFcOpen(false)}>Cancel</but
ton>
                <button type="submit" className="btn btn-primary">Save card</button>
              </div>
            </form>
          </div>
        </div>
      )}
    </>
  );
}

// Server-side rendering fetches initial data directly from SQLite!
export async function getServerSideProps() {
  try {
    const db = await getDb();
    const coursesRows = db.query("SELECT * FROM courses ORDER BY created_at ASC");
    const tasksRows = db.query("SELECT * FROM tasks ORDER BY created_at DESC");
    const flashcardsRows = db.query("SELECT * FROM flashcards ORDER BY created_at ASC");
    const pomoRows = db.query("SELECT COUNT(*) as count, COALESCE(SUM(duration_minutes), 0) as total_minutes FROM pom
odoro_sessions");

    return {
      props: {
        initialData: {
          courses: coursesRows.map(r => ({
            id: r.id,
            code: r.code,
            name: r.name,
            credits: Number(r.credits),
            grade: r.grade,
            gradePoint: Number(r.grade_point)
          })),
          tasks: tasksRows.map(r => ({
            id: r.id,
            title: r.title,
            course: r.course,
            dueDate: r.due_date,
            priority: r.priority,
            hours: Number(r.hours),
            completed: Boolean(r.completed)
          })),
          flashcards: flashcardsRows.map(r => ({
            id: r.id,
            subject: r.subject,
            question: r.question,
            answer: r.answer,
            mastered: Boolean(r.mastered)
          })),
          pomodoro: {
            completedSessions: Number(pomoRows[0]?.count || 0),
            totalMinutes: Number(pomoRows[0]?.total_minutes || 0),
            streakDays: 0
          }
        }
      }
    };
  } catch (e) {
    console.error("getServerSideProps database error:", e);
    return {
      props: {
        initialData: {
          courses: [],
          tasks: [],
          flashcards: [],
          pomodoro: { completedSessions: 0, totalMinutes: 0, streakDays: 0 }
        }
      }
    };
  }
}
```

## 6. API: pages/api/data.js (SQLite Aggregation Endpoint)

```
import { getDb } from "../../lib/db";

export default async function handler(req, res) {
  try {
    const db = await getDb();

    if (req.method === "GET") {
      const coursesRows = db.query("SELECT * FROM courses ORDER BY created_at ASC");
      const tasksRows = db.query("SELECT * FROM tasks ORDER BY created_at DESC");
      const flashcardsRows = db.query("SELECT * FROM flashcards ORDER BY created_at ASC");
      const pomoRows = db.query("SELECT COUNT(*) as count, COALESCE(SUM(duration_minutes), 0) as total_minutes FROM p
omodoro_sessions");

      const courses = coursesRows.map(r => ({
        id: r.id,
        code: r.code,
        name: r.name,
        credits: Number(r.credits),
        grade: r.grade,
        gradePoint: Number(r.grade_point)
      }));

      const tasks = tasksRows.map(r => ({
        id: r.id,
        title: r.title,
        course: r.course,
        dueDate: r.due_date,
        priority: r.priority,
        hours: Number(r.hours),
        completed: Boolean(r.completed)
      }));

      const flashcards = flashcardsRows.map(r => ({
        id: r.id,
        subject: r.subject,
        question: r.question,
        answer: r.answer,
        mastered: Boolean(r.mastered)
      }));

      const pomoCount = Number(pomoRows[0]?.count || 0);
      const pomoMinutes = Number(pomoRows[0]?.total_minutes || 0);

      return res.status(200).json({
        student: {
          name: "Student Workspace",
          role: "Academic Workspace",
          department: "Computer Science & Engineering"
        },
        courses,
        tasks,
        flashcards,
        pomodoro: {
          completedSessions: pomoCount,
          totalMinutes: pomoMinutes,
          streakDays: pomoCount > 0 ? 4 : 0
        }
      });
    }

    res.setHeader("Allow", ["GET"]);
    return res.status(405).json({ error: `Method ${req.method} not allowed` });
  } catch (err) {
    console.error("API /api/data error:", err);
    return res.status(500).json({ error: "Database error: " + err.message });
  }
}
```

## 7. API: pages/api/tasks.js (SQL CRUD for Assignments)

```
import { getDb } from "../../lib/db";

export default async function handler(req, res) {
  try {
    const db = await getDb();

    if (req.method === "POST") {
      const { title, course, dueDate, priority, hours } = req.body;
      if (!title || !course || !dueDate) {
        return res.status(400).json({ error: "Missing required fields" });
      }

      const id = "T-" + Date.now();
      const now = Date.now();

      db.run(
        "INSERT INTO tasks (id, title, course, due_date, priority, hours, completed, created_at) VALUES (?, ?, ?, ?,
?, ?, 0, ?)",
        [id, title, course, dueDate, priority || "Medium", Number(hours) || 3, now]
      );

      return res.status(201).json({ success: true, id });
    }

    if (req.method === "PATCH") {
      const { id, completed } = req.body;
      if (!id) return res.status(400).json({ error: "Task id is required" });

      db.run("UPDATE tasks SET completed = ? WHERE id = ?", [completed ? 1 : 0, id]);
      return res.status(200).json({ success: true });
    }

    if (req.method === "DELETE") {
      const { id } = req.query;
      if (!id) return res.status(400).json({ error: "Task id is required" });

      db.run("DELETE FROM tasks WHERE id = ?", [id]);
      return res.status(200).json({ success: true });
    }

    res.setHeader("Allow", ["POST", "PATCH", "DELETE"]);
    return res.status(405).json({ error: `Method ${req.method} not allowed` });
  } catch (err) {
    console.error("API /api/tasks error:", err);
    return res.status(500).json({ error: err.message });
  }
}
```

## 8. Metadata: package.json (Dependencies & Scripts)

```
{
  "name": "student-productivity-suite",
  "version": "1.0.0",
  "description": "Unified Academic Workspace with smart task analytics, SGPA forecasting, and Pomodoro study tools",
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start"
  },
  "dependencies": {
    "@libsql/client": "^0.18.0",
    "next": "^14.2.35",
    "react": "^18.3.1",
    "react-dom": "^18.3.1",
    "sql.js": "^1.16.0"
  },
  "author": "Abhishek Kumar",
  "license": "MIT"
}
```

## 9. Deployment: vercel.json (Clean URLs Configuration)

```
{
  "cleanUrls": true
}
```

# 7. Sample Exported Performance Summary

Below is an authentic sample of the plain-text academic performance report generated by the application's built-in export engine:

```
================================================================================
                    STUDENT WORKSPACE • ACADEMIC REPORT
================================================================================
Generated Date:   2026-11-12
Workspace:        Student Workspace
Current SGPA:     9.30 / 10.00
Enrolled Credits: 18.0 Credits
Study Time:       100 minutes (4 focus blocks)
Database:         SQLite WebAssembly Engine (Local Client-Side Storage)
================================================================================

1. REGISTERED SEMESTER COURSES:
--------------------------------------------------------------------------------
Code          Course Title                              Credits   Grade   Points
--------------------------------------------------------------------------------
25CSH-208     Object Oriented Programming (Java)        4.0       A+      10.0
25CSH-202     Data Structures & Algorithms              4.0       A       9.0
25CSH-205     Computer Architecture & Organization      3.0       A+      10.0
25CSH-209     Discrete Mathematical Structures          3.5       A       9.0
25CSH-212     Operating Systems & Concurrency           3.5       B+      8.0
--------------------------------------------------------------------------------

2. ASSIGNMENTS & DELIVERABLES:
--------------------------------------------------------------------------------
Title                                 Course                  Due Date     Priority
--------------------------------------------------------------------------------
[DUE]   Multithreaded Producer-Consumer  Object Oriented Progr   2026-11-18   High
[DUE]   Serialization Lab Practical      Object Oriented Progr   2026-11-20   High
[DUE]   AVL Tree Balancing Traversal     Data Structures & Algo  2026-11-25   Medium
[DONE]  MIPS Pipeline Hazard Report      Computer Architecture   2026-11-28   Low
--------------------------------------------------------------------------------

3. FLASHCARDS RETENTION:
Total Cards:      4
Mastered Topics:  2
================================================================================
```

# 8. Conclusion, Research & Bibliography

**Conclusion:** Student Workspace successfully demonstrates that an embedded local SQLite engine combined with modern web standards can deliver a superior, responsive, and completely self-contained academic management tool. By eliminating external cloud database dependencies, the application achieves instantaneous query response, complete user privacy, zero hosting costs, and effortless dual deployment on both GitHub Pages and Vercel.

**[1] Francesco Cirillo**, *The Pomodoro Technique: The Acclaimed Time-Management System That Has Transformed How We Work*, Currency / Penguin Random House, 2018.

**[2] Karpicke, J. D., & Blunt, J. R.**, *Retrieval Practice Produces More Learning than Elaborative Studying with Concept Mapping*, Science, 331(6018), 772-775.

**[3] Richard Hipp**, *SQLite Architecture and Embedded Database Engines*, SQLite Development Team.

**[4] Mozilla Developer Network**, *WebAssembly & Client-Side Storage: High-performance in-browser database execution*, MDN Web Docs, 2024.